Ministry of Surface Transport
(Roads Wing)

STANDARD PLANS FOR HIGHWAY BRIDGES
R.C.C. SLAB SUPERSTRUCTURE

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The Indian Roads Congress
on behalf of the Govt. of India,
Ministry of Surface Transport (Roads Wing)

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FOREWORD

I have great pleasure in placing before the community of Bridge Engineers, this Volume of Standard Plans for Highway Bridges covering RCC right slab type superstructure. This publication is the first in the series of new Standard Plans for Highway Bridge Superstructure being brought out keeping in view the recent changes in specifications and provisions in the Bridge Codes. I am sure this publication will prove extremely useful in proper planning, estimation and execution of highway bridges in the country.

The publication has been made possible by the sustained efforts of the personnel of the Bridges Standards and Research zone of the Bridges Directorate of this Ministry and the Consultant associated with the work, who deserve commendation for the work done by them. The keen interest of the Addl. Director General (Bridges), in taking up this work and bringing out this publication in a short time is worthy of special mention.

(K.K. SARIN)
Director General (Road Development) &
Addl. Secretary to the Govt. of India

New Delhi, June 1, 1991

PREFACE

The Standard Plans for Highway Bridges with RCC Slab Type Superstructure (Volume II) were first issued by the Ministry of Surface Transport (Roads Wing) in the year 1977. Since then there have been several revisions in the specifications and provisions of the Bridge Codes. The preparation of revised Standard Drawings was, therefore, taken up towards the end of 1989 and completed on top priority.

This Volume, the first in the series of new Standard Drawings for Superstructure, contains Standard Plans for RCC Slab type highway bridge superstructure for 3.0 to 10.0 metre effective spans. It also contains drawings for wearing coat, railings and miscellaneous items. A separate volume containing bill of quantities for various items of superstructures will also be issued shortly.

The design caters for one lane of IRC Class 70-R wheeled/tracked loading or 2-lanes of IRC Class A loading whichever produces more severe effect. Footpaths have been designed for a crowd load of 5 kN/m^2. Keeping in view the current practice of providing a deck of the same width as the adjoining road for NH bridges having total length less than 30 m, the overall width between the outer faces of the railing kerb has been kept as 12 m. The wearing coat will be of mastic and asphaltic concrete type, except in remote areas where average 75 mm thick cement concrete wearing coat may also be adopted. The designs are based on Standard Specifications and Codes of Practice for Highway Bridges issued by the Indian Roads Congress. For construction purposes, Specifications for Road and Bridge works issued by the Govt. of India, Ministry of Surface Transport (Roads Wing), as amended from time to time, will apply.

The plans have been made complete in all respects so that they could be readily adopted for preparation of estimates and also serve as construction drawings in the field. The entire design philosophy adopted lays great emphasis on constructability i.e. convenient and full translation of the design on to the ground. A great deal of attention has, therefore, been paid to dimensioning and detailing. I have no doubt that the wide spread adoption of these Standard Plans will lead to reduction in time of construction and enhancement of the quality and durability of our road bridges.

Every possible care has been taken to eliminate errors in the Drawings but users are requested to bring to our notice errors or omissions, if any, which may come to light while using these Drawings in their bridge works.

The work of preparing the Designs and Drawings was carried out by the Consultant, M/s. Consulting Engineering Services (India) Pvt. Ltd., New Delhi. Equally important contributions in the finalisation of the designs and details were made by officers of the Ministry whose names appear in the title blocks of various drawings. The enthusiasm and dedication which they brought to bear on the task are to be highly appreciated.

(NINAN KOSHI)
Addl. Director General (Bridges),
Ministry of Surface Transport (Roads Wing)

New Delhi, June 1, 1991

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	"	10.0 m	SD/122

(A) GENERAL

1. These notes are applicable for the Standard Drawings for R.C.C. solid slab superstructure with and without footpaths.
2. These drawings are applicable only for right bridges with overall width of 12 m.
3. No raised footpaths shall be provided on the bridges having length less than 30m unless the same are otherwise existing on the approaches.
4. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed. No drawing shall be scaled.
5. Design criteria:
   - I. The design is according to the following codes:
     - (a) IRC : 5-1985
     - (b) IRC : 6-1966 (1985 reprint)
     - (c) IRC : 21-1987.
   - II. The following loads have been considered in the design:
     - (a) One lane of IRC class 70R or two lanes of IRC class A on carriage way, whichever governs.
     - (b) Footpath load of 5 kN/sq.m for superstructure having footpaths.
     - (c) Wearing coat load of 2 kN/sq.m.
   - III. The designs are applicable for 'MODERATE' AND 'SEVERE' conditions of exposure.
6. Public utility services (except water supply and sewerage), if required, shall be carried over the bridge through 150mm diameter ducts provided in the footpaths. Total load of such services shall not be more than 1.0 kN per metre on each footpath. Water/sewerage pipeline shall not be carried over any part of the superstructure. Inspection chambers in footpaths may be provided as shown in the drawing. The location and spacing of chambers along the footpath will be decided by the Engineer-in-charge in consultation with the users.
7. Wearing coat shall consist of the following:
   - I.(a) A coat of mastic asphalt 6mm thick with a prime coat over the top of the deck before the wearing coat is laid. The prime coat of mastic asphalt shall be 30% straight run 30/40 penetration grade bitumen and 50% light solvent (Benzol) to be laid over the deck slab. The insulating layer of 6mm thick mastic asphalt with 75% lime stone dust filler and 25% of 30/40 penetration grade bitumen shall be laid at 375°F with broom over prime coat.
     - (b) 50mm thick asphaltic concrete wearing coat in two layers of 25mm each as per Clause 512 of MOST's Specifications for Road and Bridge Works (Second Revision-1988)
   - II. In case of isolated bridge construction or bridges located in remote areas where provision of mastic and asphaltic concrete wearing coat is not practicable, the Engineer-in-charge may permit provision of 75mm thick cement concrete wearing coat in M30 grade concrete with maximum water cement ratio as 0.40. The reinforcement shall consist of 8mm High Yield Strength Deformed bars @ 200mm centres reducing to 100 centres in both the directions over a strip of 300mm near the expansion joint. Reinforcement shall be placed at the centre of the wearing coat. Wearing coat shall be discontinued at expansion joint locations. Joint fillers shall extend upto the top of wearing coat.
8. 20mm expansion joint does not cater for any allowance for possible tilting of abutment.
9. Support for the deck slab shall provide a bearing width of 400mm.
10. In urban areas, chequered tiles may be provided in the footpath portion by suitably adjusting the thickness of the footpath slab.
11. Type/position of return walls, railings, guard posts, ramp etc. in approach portion shall be decided by the Engineer-in-charge.

(B) MATERIALS SPECIFICATIONS

Concrete

1. Concrete shall be of design mix and shall have minimum 28 days characteristic strength on 150mm cubes for all elements of superstructure as indicated below:

Conditions of exposure	Concrete grade	Characteristic Strength
'MODERATE'	M 25	25 MPa (for 3m to 9m span)
'MODERATE'	M 30	30 MPa (for 10m span)
'SEVERE'	M 30	30 MPa (for 3m to 10m span)

2. High strength ordinary portland cement conforming to IS:8112 or ordinary portland cement conforming to IS 269 capable of achieving the required design concrete strength shall only be used.
3. The minimum cement content and maximum water cement ratio in the concrete design mix shall be 310 kg/cu.m and 0.45 respectively for 'MODERATE' conditions of exposure. The minimum cement content and maximum water cement ratio in the concrete design mix shall be 400 kg/cu.m and 0.40 respectively for 'SEVERE' conditions of exposure.

Reinforcement

All reinforcing bars shall be High Yield Strength Deformed bars (Grade designation S 415) conforming to IS 1786.

Water

Water to be used in concreting and curing shall conform to Clause 302.4 of IRC 21-1987.

(C) WORKMANSHIP/DETAILING

1. Minimum clear cover to any reinforcement including stirrups shall be 50mm unless shown otherwise in the drawings.
2. For ensuring proper cover of concrete to reinforcement bars specially made polymer cover blocks shall only be used.
3. **Construction Joints**
   - I. The location and provision of construction joints shall be approved by Engineer-in-charge. The concreting operation shall be carried out continuously upto the construction joint.
   - II. The concrete surface at the joint shall be brushed with a stiff brush after casting while the concrete is still fresh and it has only slightly hardened.
   - III. Before new concrete is poured the surface of old concrete shall be prepared as under:
     - a) For hardened concrete, the surface shall be thoroughly cleaned to remove debris/laitance and made rough so that 1/4 of the size of the aggregate or structurally damaging the concrete.
     - b) For partially hardened concrete, the surface shall be treated by wire brush followed by an air jet.
     - c) The old surface shall be soaked with water without leaving puddles immediately before starting concreting to prevent the absorption of water from new concrete.
   - IV. New concrete shall be thoroughly compacted in the region of the joint.
4. Welding of reinforcement bars shall not be permitted.
5. Lape in reinforcement:
   - I. Minimum lap length of reinforcement shall be kept as 83 d where 'd' is the diameter of bar.
   - II. Not more than 50% of reinforcement shall be lapped at any one location.
6. Bending of reinforcement bars shall be as per IS : 2502.
7. Supporting chairs of 12mm diameter shall be provided at suitable intervals as per IS : 2502.
8. Concrete shall be produced in a mechanical mixer of capacity not less than 200 litres having integral weigh-batching facility and automatic water measuring and dispensing device.
9. Proper compaction of concrete shall be ensured by use of full width screed vibrators for concrete in deck slab.
10. Properly braced steel plates shall be used as shuttering.
11. Sharp edges of concrete shall be chamfered.

(D) GENERAL SPECIFICATIONS

The work shall be executed in accordance with MOST's Specification for Road and Bridge Works (Second Revision, 1988) except wherever otherwise mentioned.

(E) REFERENCE TO DRAWINGS

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SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD//105	DETAILS OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)
SD/106	DETAILS OF R.C.C. RAILINGS (WITH FOOTPATHS)
SD/107 THROUGH SD/114	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3m To 10m (WITHOUT FOOTPATHS)
SD/115 THROUGH SD/122	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3m To 10m (WITH FOOTPATHS)

MKD	DATE	DESCRIPTION	BY

REVISION

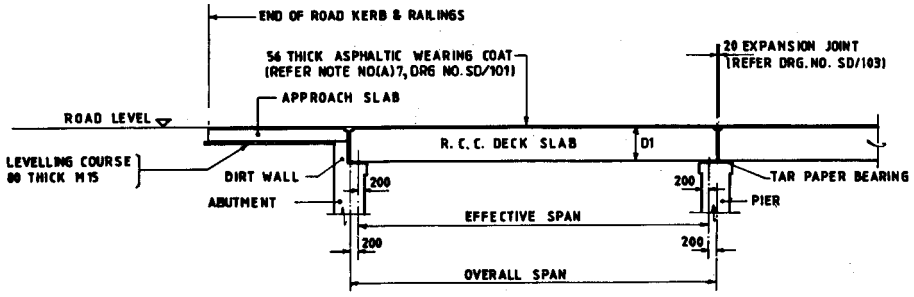
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTUCTURE (RIGHT) SPAN 3.0m To 10.0 m
(WITH AND WITHOUT FOOTPATHS)
GENERAL NOTES

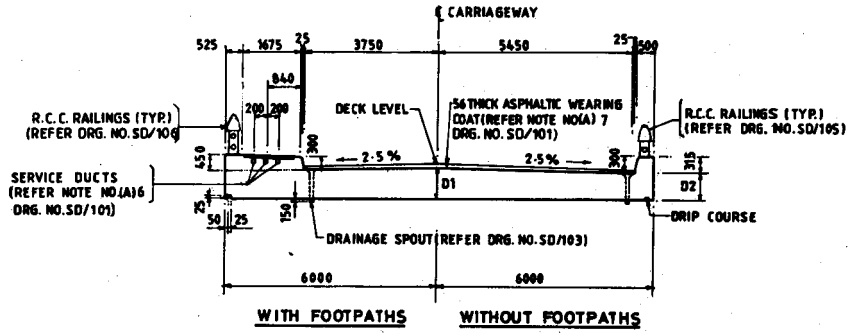
RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K MUKHERJEE) C.E.	DRG. NO. SD/101

DRG. NO. SD/102



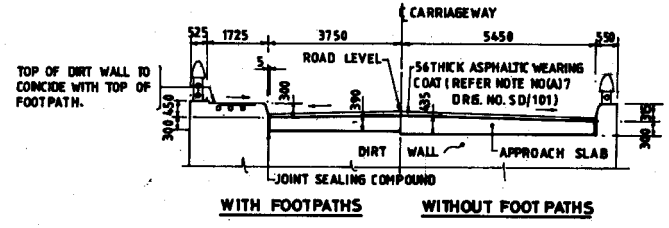
SECTION. A-A

(ROAD KERB & RAILINGS NOT SHOWN.)

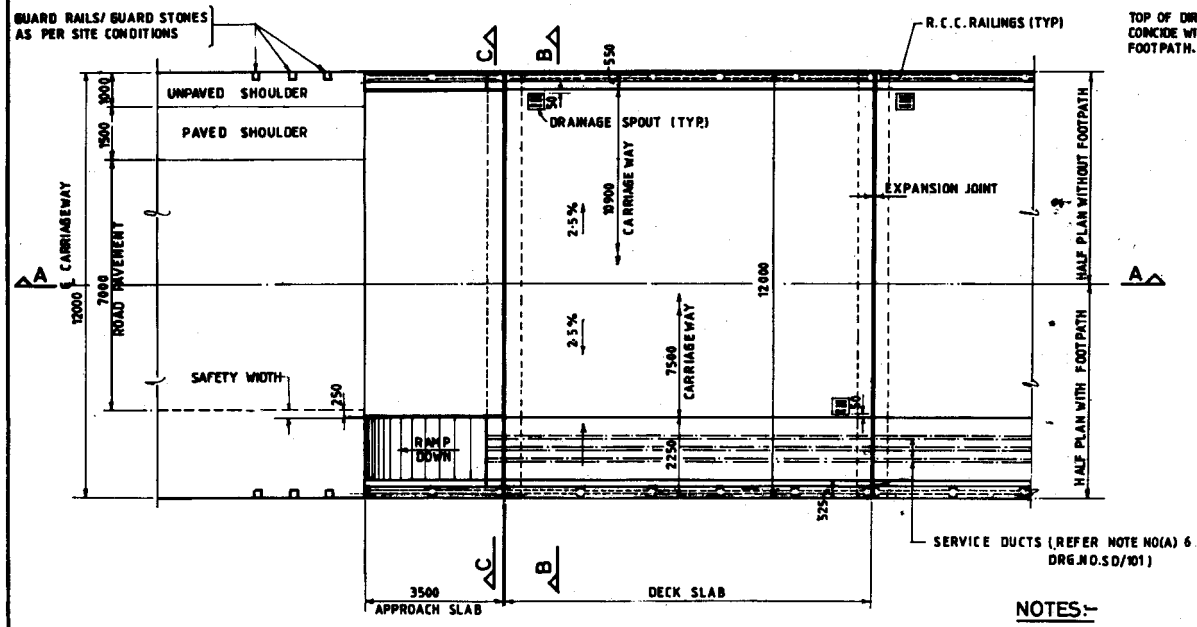


SECTION B-B

FOR VALUES OF D1 & D2 REFER RELEVANT DRAWINGS (DRG. NO. SD/107 THROUGH SD/122.)



SECTION. C-C



PLAN.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)
SD/106	DETAILS OF R.C.C. RAILINGS (WITH FOOTPATHS)
SD/107 THROUGH SD/114	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3·0 m TO 10·0 m (WITHOUT FOOTPATHS)
SD/115 THROUGH SD/122	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3·0 m TO 10·0 m (WITH FOOTPATHS)

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Typical arrangement of drainage spouts has been shown in plan. Suitable modifications may be made by the Engineer-in-charge as per site conditions and intensity of rainfall.

MKD	DATE	DESCRIPTION	BY

R E V I S I O N

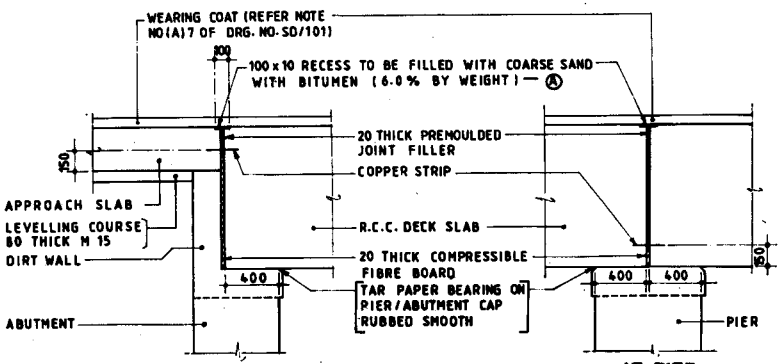
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 3·0m TO 10·0m
(WITH AND WITHOUT FOOTPATHS)
GENERAL ARRANGEMENT

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/102

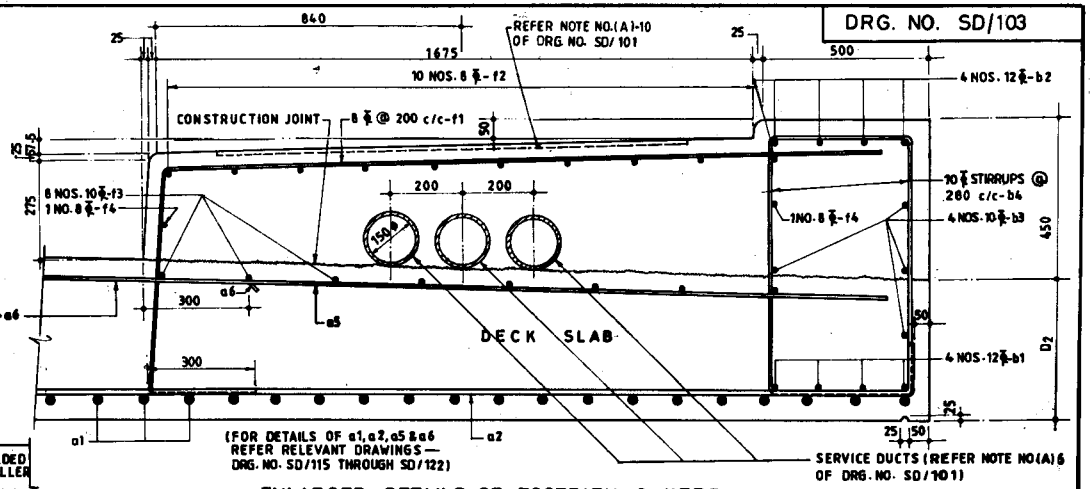
DRG. NO. SD/103



AT ABUTMENT

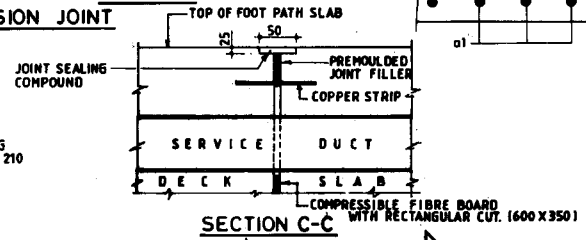
AT PIER

DETAILS OF EXPANSION JOINT

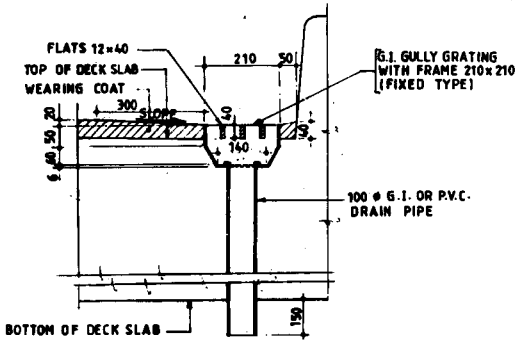


ENLARGED DETAILS OF FOOTPATH & KERB

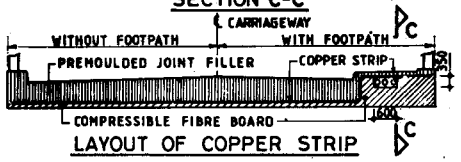
(REINFORCEMENT OF RAILING POST NOT SHOWN)



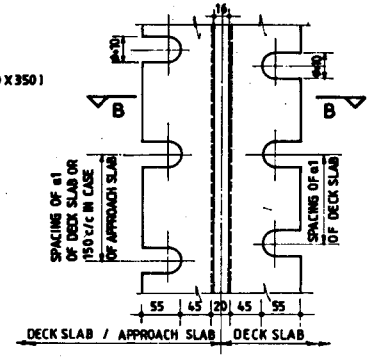
SECTION C-C



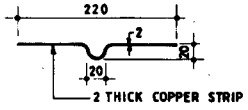
SECTION A-A



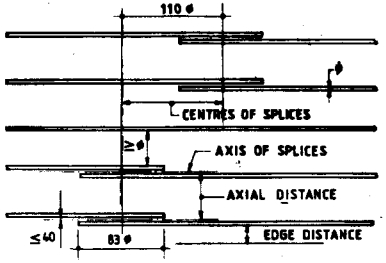
LAYOUT OF COPPER STRIP



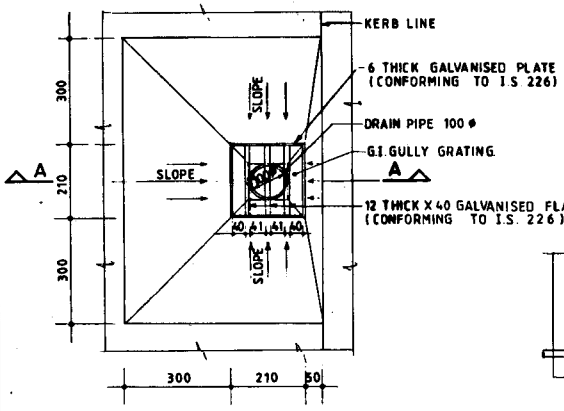
DETAIL OF COPPER STRIP



SECTION B-B

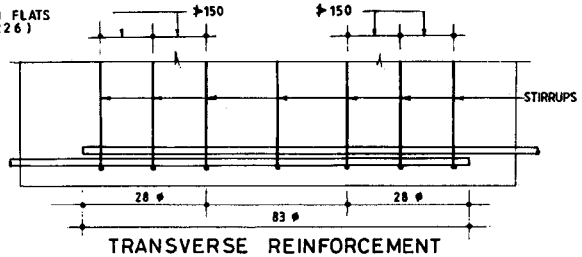


LAP SPLICES OF BARS IN TENSION



PLAN

(DETAILS OF DRAINAGE SPOUT & COLLECTION PIT)



TRANSVERSE REINFORCEMENT AT LAP SPLICES IN KERBS

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/104	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R. C. C. RAILINGS (WITHOUT FOOTPATHS)
SD/106	DETAILS OF R. C. C. RAILINGS (WITH FOOTPATHS)
SD/107 THROUGH SD/114	R.C.C SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3.0m TO 10.0m (WITHOUT FOOTPATHS)
SD/115 THROUGH SD/122	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3.0m TO 10.0m (WITH FOOTPATHS)

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Splicing of main longitudinal bars in deck slab shall be avoided as far as practicable. In case the splicing of bars becomes unavoidable, the arrangement shall be as shown in the drawing.
3. Area of bars spliced at any section shall not exceed 50 % of the total area of the bars provided at the section.
4. Joint filler shall conform to IS 1838. Products with I.S.I certification mark shall only be used.
5. The reinforcement of deck slab shall be suitably modified to accommodate the drainage spout. Also refer note no. 2 of drawing no. SD/102. The reinforcements marked f1 and f2 shall be modified in the region of inspection chamber.
6. The drainage spout shall be galvanised after welding the plates/flats.

MKD	DATE	DESCRIPTION	BY

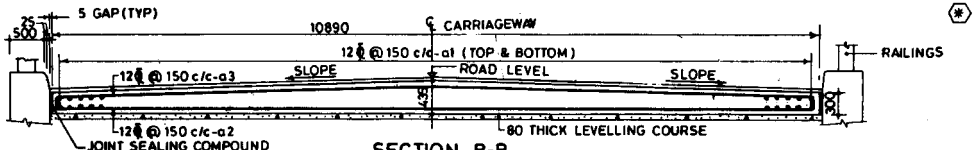
REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

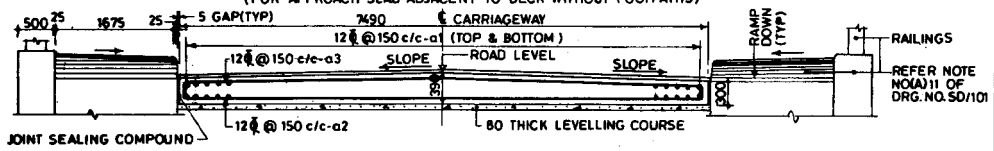
R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 3.0m TO 10.0m
(WITH AND WITHOUT FOOTPATHS)
MISCELLANEOUS DETAILS

RECOMMENDED BY		APPROVED BY	1990
(S.S. REDDY) E.E.	(S.K. KAISTHA) S.E.	(P.N. SHIVAPRASAD) C.E.	DRG. NO. SD/103



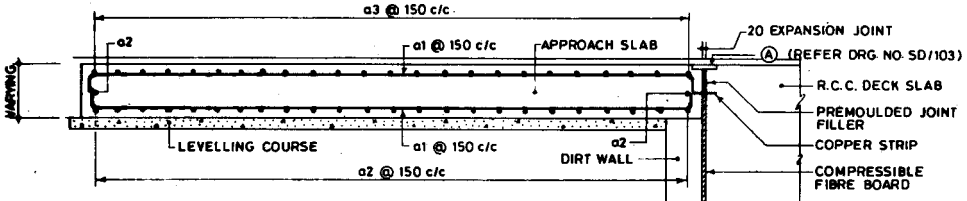
SECTION B-B

(FOR APPROACH SLAB ADJACENT TO DECK WITHOUT FOOTPATHS)

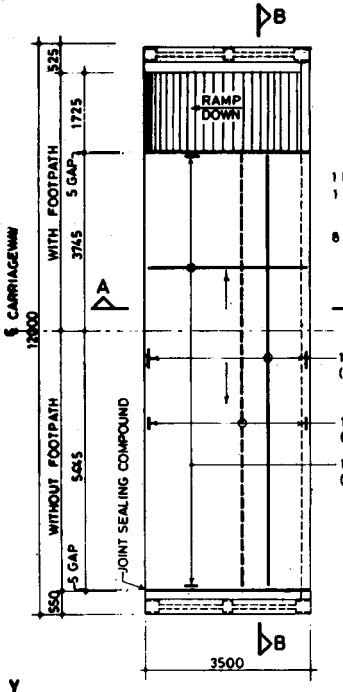


SECTION B-B

(FOR APPROACH SLAB ADJACENT TO DECK WITH FOOTPATHS)

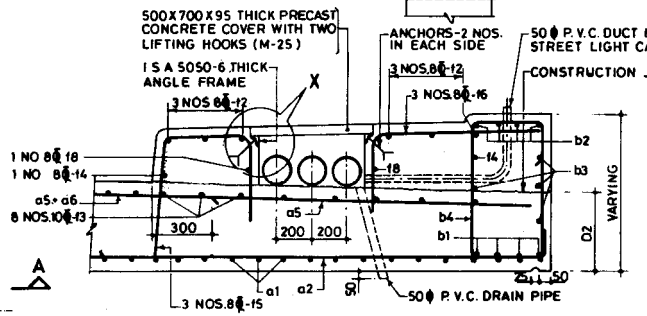


SECTION A-A



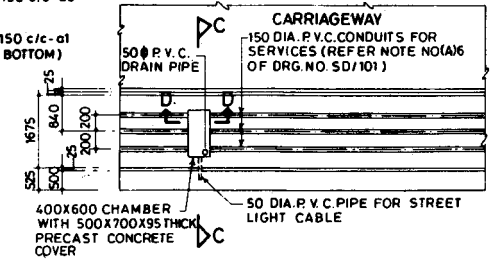
PLAN

(SHOWING REINFORCEMENT DETAILS OF APPROACH SLAB)
(ABUTMENT, DIRTWALL & LEVELLING COURSE NOT SHOWN)



SECTION C-C

(REINFORCEMENT OF RAILING POST NOT SHOWN
FOR VALUES OF D2 & DETAILS OF a1, a2, a5, a6, b1, b2, b3 & b4
REFER RELEVANT DRAWINGS :
– DRG. NOS. SD/115 THROUGH SD/122)



PART PLAN OF BRIDGE DECK

(SHOWING DETAILS OF INSPECTION CHAMBER IN FOOTPATH
THE LOCATION AND SPACINGS ALONG THE FOOTPATHS TO BE
DECIDED BY ENGINEER-IN-CHARGE AS PER SITE CONDITIONS
AND IN CONSULTATION WITH THE CONCERNED AUTHORITIES)

⊛ SCHEDULE OF REINFORCEMENT (PER APPROACH SLAB)

DRG. NO. SD/104

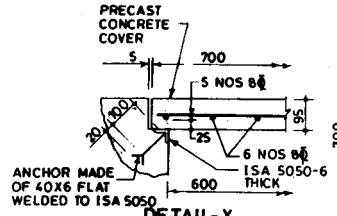
SL.NO.	LOCATION	BAR MKD.	SHAPE	DIA. (m.m.)	SPACING (m.m.)	LENGTH (m.m.)	NOS.	TOTAL LENGTH (m)	WEIGHT (kg)
APPROACH SLAB ADJACENT TO DECK WITHOUT FOOTPATHS									
1	SLAB TOP & BOTTOM ALONG-X	a1	3400; 200; R=50; 200	12	150	3730	146	544·58	484
2	SLAB BOTTOM ALONG-Y	a2	150; R=50; 10790; 150	12	150	11020	26	286·52	254
3	SLAB TOP ALONG-Y	a3	5395 + 5395; 150; R=50; 135; 150	12	150	11020	24	264·48	235
APPROACH SLAB ADJACENT TO DECK WITH FOOTPATHS									
4	SLAB TOP & BOTTOM ALONG-X	a1	3400; 200; R=50; 200	12	150	3730	100	373·00	331
5	SLAB BOTTOM ALONG-Y	a2	150; R=50; 7390; 150	12	150	7620	26	198·12	176
6	SLAB TOP ALONG-Y	a3	3695 + 3695; 150; R=50; 90; 150	12	150	7620	24	182·88	162

QUANTITIES

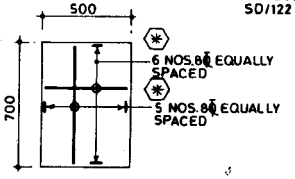
CONCRETE IN APPROACH SLAB (cu.m.)	WITH FOOTPATHS	9·04
	WITHOUT FOOTPATHS	14·01
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	WITH FOOTPATHS	702
	WITHOUT FOOTPATHS	1022
ASPHALTIC WEARING COAT (sq.m.)	WITH FOOTPATHS	26·22
	WITHOUT FOOTPATHS	38·12

REFERENCE DRAWINGS

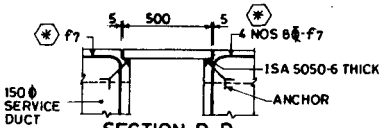
DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)
SD/106	DETAILS OF R.C.C. RAILINGS (WITH FOOTPATHS)
SD/107 THROUGH SD/114	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3·0m TO 10·0m (WITHOUT FOOTPATHS)
SD/115 THROUGH SD/122	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3·0m TO 10·0m (WITH FOOTPATHS)



DETAIL-X



DETAILS OF PRECAST CONCRETE COVER FOR INSPECTION CHAMBER



SECTION D-D

(SHOWING SHAPE OF BARS IN THE REGION OF INSPECTION CHAMBER
OTHER REINFORCEMENTS ARE NOT SHOWN)

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. The reinforcement of railing posts shall be incorporated before casting of deck slab. The railings shall conform to drawing no. SD/105 or SD/106 or any other approved type.
3. Dimensions in schedule of reinforcement are given as per I.S. 2502
4. ⊛ Schedule of reinforcement does not include change in reinforcement around inspection chamber, precast cover to chamber and railings.

MKD	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

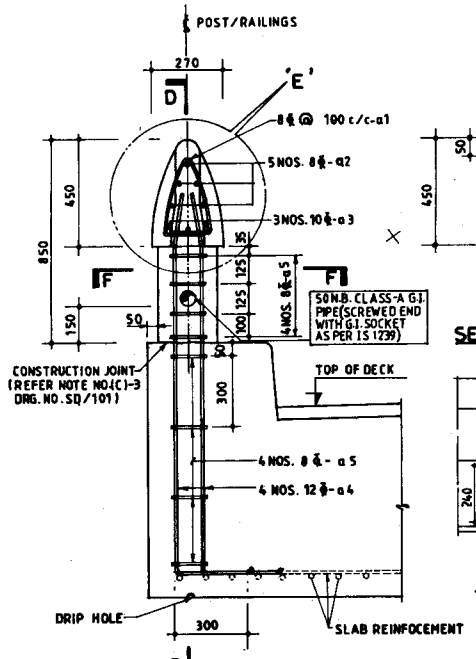
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 3·0m TO 10·0m
(WITH AND WITHOUT FOOTPATHS)

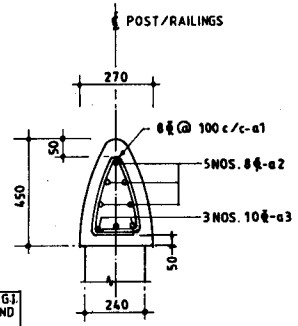
MISCELLANEOUS DETAILS

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S. K. KAISTHA) S.E.	(P. N. SHIVAPRASAD) C.E.	DRG. NO. SD/104

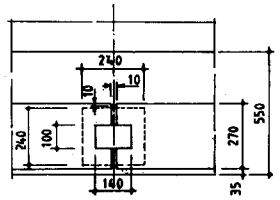
DRG. NO. SD/105



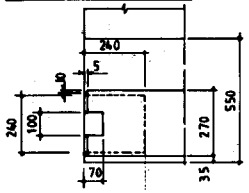
SECTION A - A



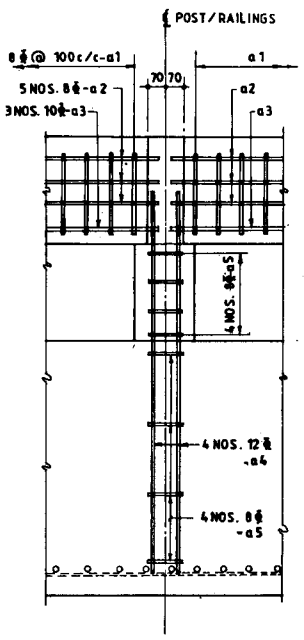
SECTION B-B



PART PLAN C-C



PART PLAN G-G



SECTION D - D

SCHEDULE OF REINFORCEMENT FOR RAILING ON ONE SIDE

SL. NO.	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS	TOTAL LENGTH(m)	WEIGHT (kg)
1	RAILINGS	a1		8	100	940	$\left[\frac{A-160}{100}+1\right]=N1$	$N1\times\frac{940}{1000}=L1$	L1 × 0·395
2	RAILINGS	a2		8	-	(A-50)	5 × N = N2	$N2\times\frac{(A-50)}{1000}=L2$	L2 × 0·395
3	RAILINGS	a3		10	-	(A-50)	3 × N = N3	$N3\times\frac{(A-50)}{1000}=L3$	L3 × 0·617
4	POST	a4	225, h, 300; 225 *, h, 300	12	-	h + 489	4 × (N + 1) = N4	$\frac{h+489}{1000}\times N4=L4$	L4 × 0·888
5	POST	a5	124, 124	8		688	8 × (N + 1) = N5	$\frac{688}{1000}\times N5=L5$	L5 × 0·395

* FOR END POST INNER TWO BARS THE SHAPE IS h, 300

QUANTITIES (PER SPAN)

EFFECTIVE SPAN(m)	NUMBER OF DIVISIONS 'N'	A (mm)	h (mm)	QUANTITY STEEL (Kg) ●	CONCRETE (m^3) RAILINGS	CONCRETE (m^3) RAIL POSTS
3	2	1700	915	95	0·65	0·14
4	3	1467	965	127	0·84	0·18
5	3	1800	1015	143	1·03	0·18
6	4	1600	1065	176	1·22	0·23
7	4	1850	1115	195	1·41	0·23
8	5	1680	1215	230	1·60	0·27
9	5	1880	1285	249	1·79	0·27
10	6	1734	1365	288	1·98	0·32

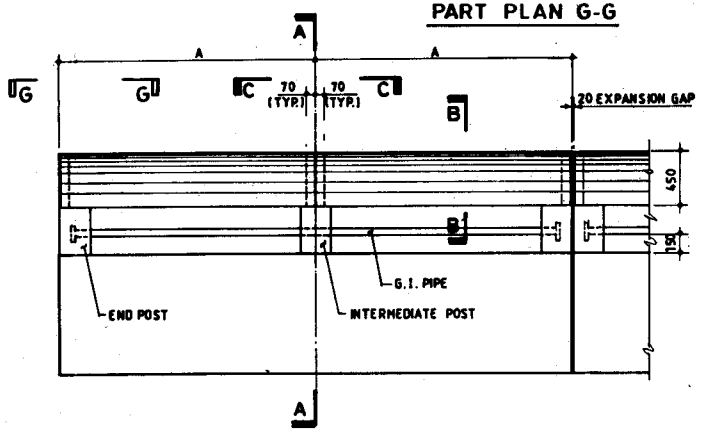
● STEEL QUANTITIES INCLUDE 5% EXTRA FOR LAPS & WASTAGE. THE QUANTITIES ARE FOR RAILINGS ON BOTH SIDES OF THE SLABS.

NOTES:-

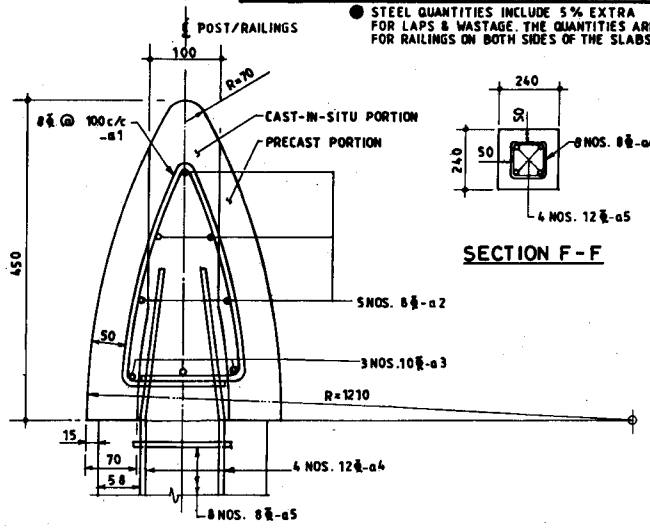
1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. The maximum size of the aggregate to be used for precast concrete work shall be limited to 12mm.
3. The centre to centre spacing 'A' between successive vertical posts shown in the elevation shall be adjusted to suit the length of the bridge span for which the railing is used but in no case shall its value exceed 2000.
4. The details of reinforcement in kerb and deck slab have not been shown in the drawing.
5. Railings to be built after the structural concrete of superstructure has hardened and the shuttering for the superstructure is released.
6. Railings shall be carefully constructed true to line and grade. Post shall be vertical with a tolerance not to exceed 1 in 500.
7. Special care shall be taken to match the surface of the cast-in-situ portion.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/107 THROUGH SD/114	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3·0m TO 10·0m (WITHOUT FOOTPATHS)



GENERAL ELEVATION



DETAIL - E

SECTION F - F

MKD	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

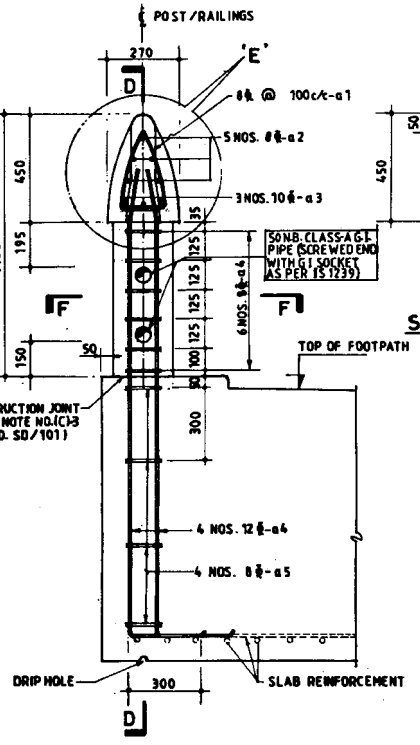
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 3·0m TO 10·0m
(WITHOUT FOOTPATHS)

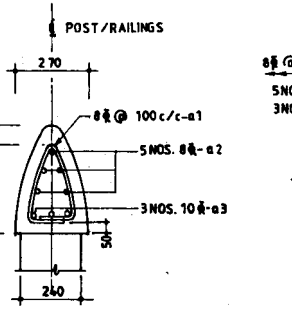
DETAILS OF R.C.C. RAILINGS

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/105

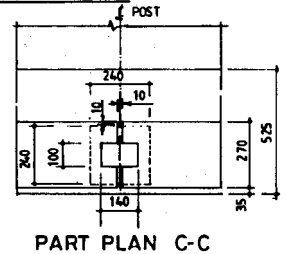
DRG. NO. SD/106



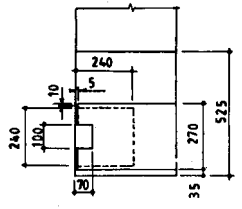
SECTION A-A



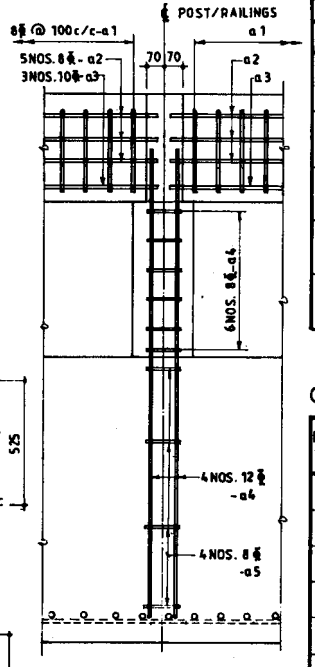
SECTION B-B



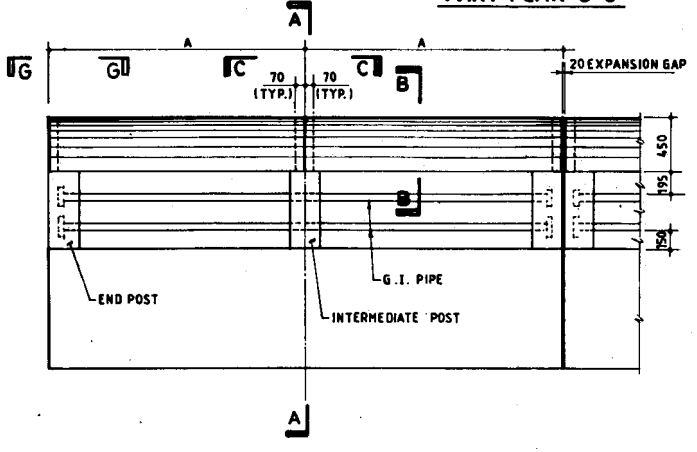
PART PLAN C-C



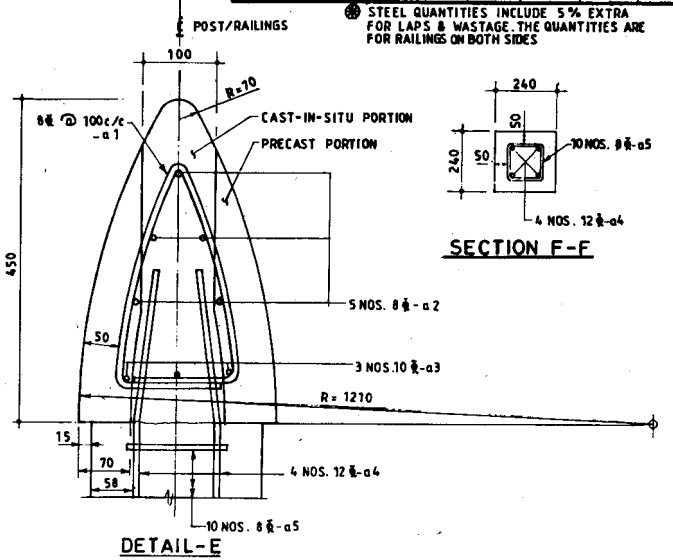
PART PLAN G-G



SECTION D-D



GENERAL ELEVATION



DETAIL-E

SECTION F-F

SCHEDULE OF REINFORCEMENT FOR RAILING ON ONE SIDE

SL. NO	LOCATION	BAR MKD	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS	TOTAL LENGTH (m)	WEIGHT kg
1	RAILINGS	a1		8	100	940	$\left[\frac{A-160}{100}+1\right]\times N$ & N1	$N1 \times \frac{940}{1000} \simeq L1$	L1 × 0·395
2	RAILINGS	a2		8	-	(A-50)	7 × N = N2	$\frac{N2 \times (A-50)}{1000} \simeq L2$	L2 × 0·395
3	RAILINGS	a3		10	-	(A-50)	3 × N = N3	$\frac{N3 \times (A-50)}{1000} \simeq L3$	L3 × 0·617
4	POST	a4	225, h, 300; 225*, h, 300	12	-	h + 489	4 × (N+1) = N4	$\frac{h+489 \times N4}{1000} \simeq L4$	L4 × 0·888
5	POST	a5	124, 124	8	-	688	10 × (N+1) = N5	$\frac{688 \times N5}{1000} \simeq L5$	L5 × 0·395

* FOR END POST INNER TWO BARS, THE SHAPE IS h, 300

QUANTITIES (PER SPAN)

EFFECTIVE SPAN(m)	NUMBER OF DIVISIONS 'N'	A (mm)	h (mm)	QUANTITY STEEL (Kg) ⊕	CONCRETE (m³) RAILINGS	CONCRETE (m³) RAIL POSTS
3	2	1700	1300	107	0·65	0·22
4	3	1467	1350	143	0·84	0·29
5	3	1800	1400	159	1·03	0·29
6	4	1600	1450	196	1·22	0·36
7	4	1850	1500	215	1·41	0·36
8	5	1680	1600	254	1·60	0·44
9	5	1880	1670	273	1·79	0·44
10	6	1734	1750	316	1·98	0·51

⊕ STEEL QUANTITIES INCLUDE 5% EXTRA FOR LAPS & WASTAGE. THE QUANTITIES ARE FOR RAILINGS ON BOTH SIDES

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. The maximum size of the aggregate to be used for precast concrete work shall be limited to 12 mm.
3. The centre to centre spacing 'A' between successive vertical posts shown in the elevation shall be adjusted to suit the length of the bridge span for which the railing is used but in no case shall its value exceed 2000.
4. The details of reinforcement in kerb and deck slab have not been shown in the drawing.
5. Railings to be built after the structural concrete of superstructure has hardened and the shuttering for the superstructure is released.
6. Railings shall be carefully constructed true to line and grade. Post shall be vertical with a tolerance not to exceed 1 in 500.
7. Special care shall be taken to match the surface of the cast-in-situ portion.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/115 THROUGH SD/122	R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPANS 3·0m TO 10·0m (WITH FOOTPATHS)

MKD	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

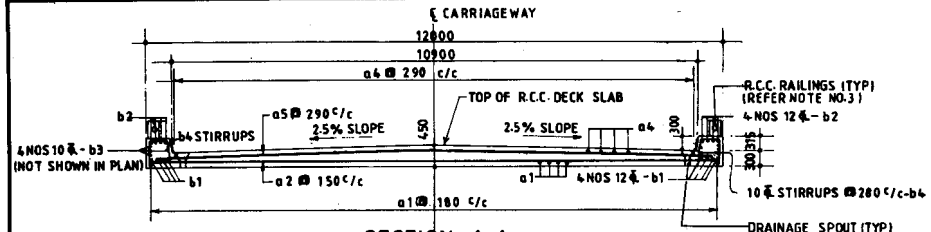
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 3·0m TO 10·0m
(WITH FOOTPATHS)

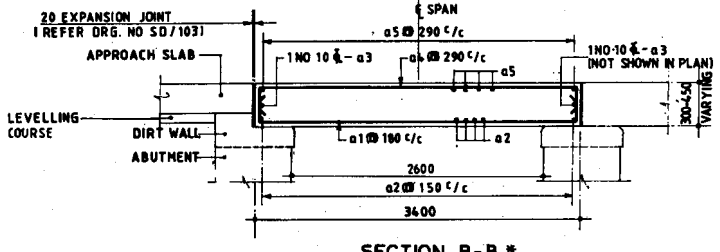
DETAILS OF R.C.C. RAILINGS

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/106

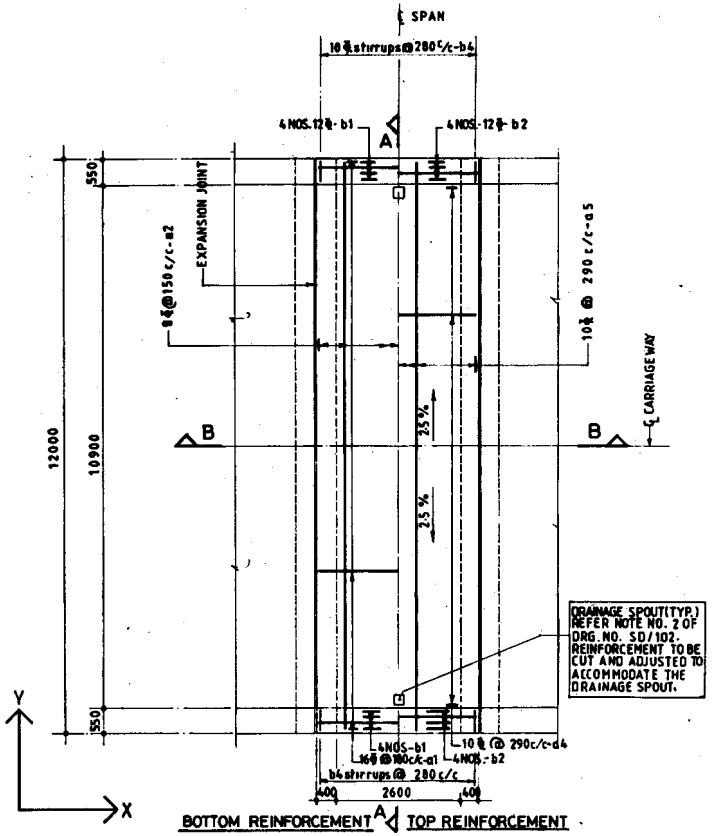
DRG. NO. SD/107



SECTION A-A



SECTION B-B *



BOTTOM REINFORCEMENT A TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	200 R=60 3300 800	16	180	3600	67	241.20	381
2	SLAB-BOTTOM ALONG-Y	a2	200 R=30 11900 200	8	150	12250	23	281.75	111
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	3300 200 R=40 200	10	290	3640	39	141.96	88
5	SLAB-TOP ALONG-Y	a5	5800 5800	10	290	11600	13	150.80	93
6	KERB-BOTTOM	b1	350 R=50 3292 350	12	—	3930	8	31.36	28
7	KERB-TOP	b2	3300 350 R=50 350	12	—	3930	8	31.44	28
8	KERB-SIDES	b3	200 R=40 3260 200	10	—	3810	8	30.48	19
9	KERB-STIRRUPS	b4	270 150 200 430	10	280 ●	1995	26 ●	51.74	32

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS - REFER DRG. NO. SD/103.

Dead load of the superstructure per span including R.C.C. railings @ 3kN/m and wearing coat @ 2 kN/sq.m. = 489 kN.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/105	DETAIL OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	3.0
CONCRETE (cu.m.)	16.42
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	835
ASPHALTIC WEARING COAT (sq.m)	37.06

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
3. Special attention is invited to note no (B)-1 of drawing no. SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502.
* 5. Reinforcement of adjacent span superstructure, approach slab not shown.

MKD	DATE	DESCRIPTION	BY

REVISION

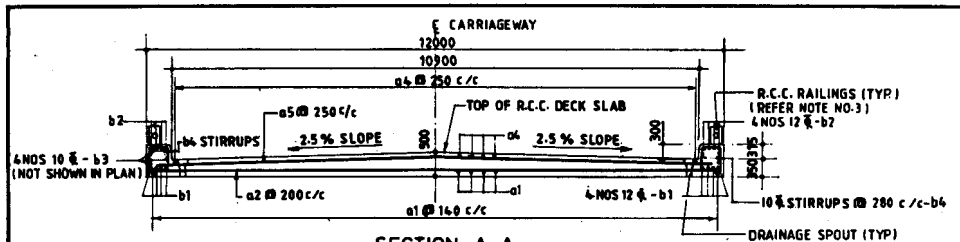
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

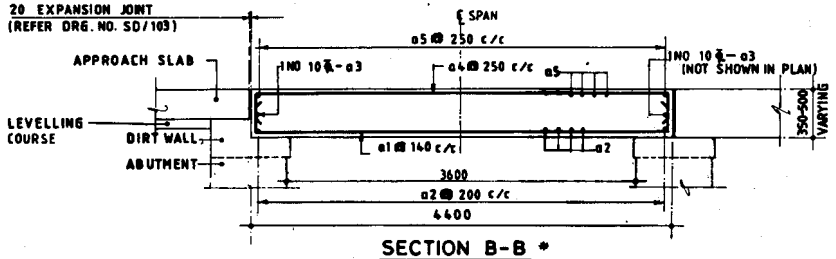
R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 3.0m
(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/107

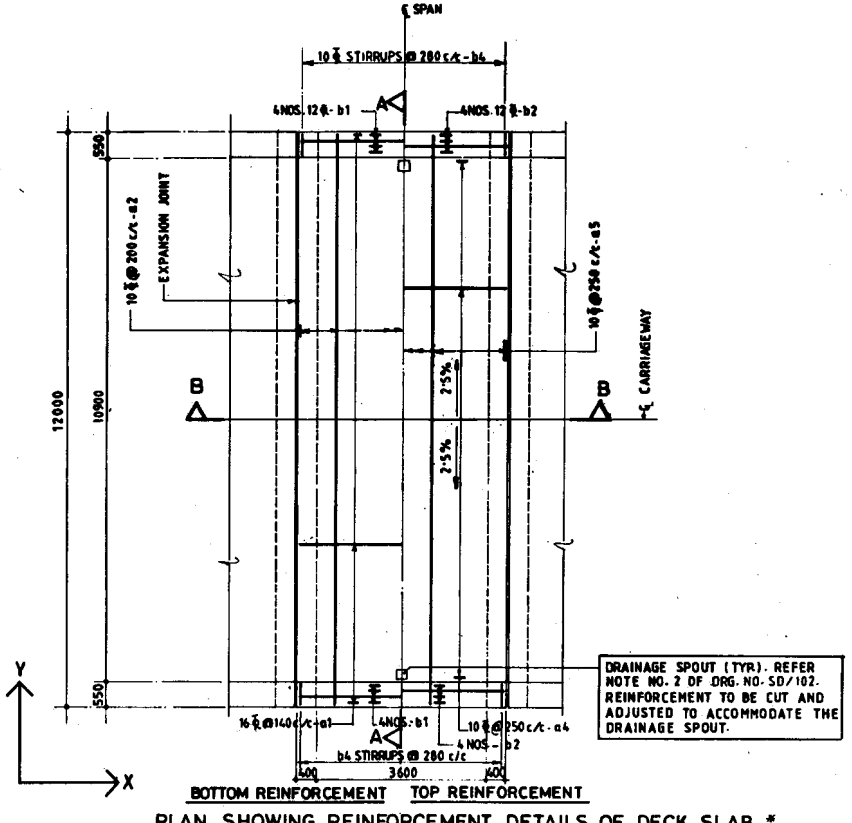
DRG. NO. SD/108



SECTION A-A



SECTION B-B *



BOTTOM REINFORCEMENT TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL NO	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH (m)	WEIGHT (kg.)
1	SLAB-BOTTOM ALONG - X	a1	250, R=60, 4300, 250	16	140	4700	86	404.20	639
2	SLAB-BOTTOM ALONG - Y	a2	200, R=40, 11900, 200	10	200	12240	23	281.52	174
3	SLAB-SIDE ALONG - Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG - X	a4	4300, 250, R=40, 250	10	250	4740	45	213.30	132
5	SLAB-TOP ALONG - Y	a5	5800, 5800, 400	10	250	11600	19	220.40	136
6	KERB - BOTTOM	b1	350, R=50, 4292, 350	12	—	4920	8	39.36	35
7	KERB - TOP	b2	4300, 250, R=50, 250	12	—	4930	8	39.44	35
8	KERB - SIDES	b3	225, R=40, 4268, 225	10	—	4860	8	38.88	24
9	KERB - STIRRUPS	b4	254, 405, 320, 430	10	280●	2100	34●	71.40	44

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS-REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3kN/m and wearing coat @ 2 kN/sq. m. = 696 kN

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	4.0
CONCRETE (cu.m.)	23.89
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	1296
ASPHALTIC WEARING COAT (sq.m)	47.96

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no (B)-1 of drawing no. SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502.

* 5 Reinforcement of adjacent span superstructure, approach slab not shown.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)

MKD	DATE	DESCRIPTION	BY

REVISION

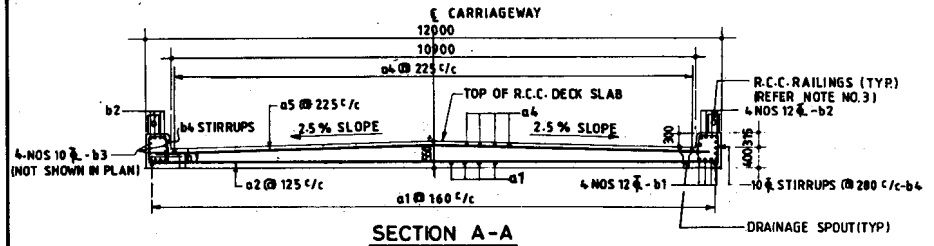
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

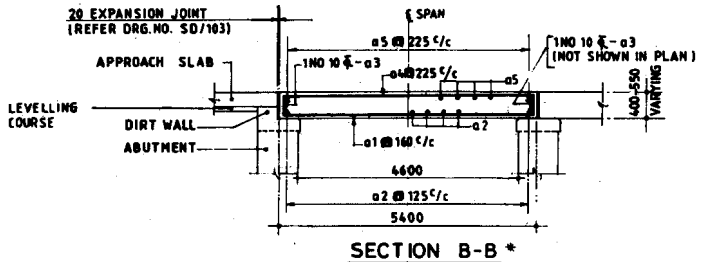
R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 4.0m
(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/108

DRG. NO. SD/109



SECTION A-A



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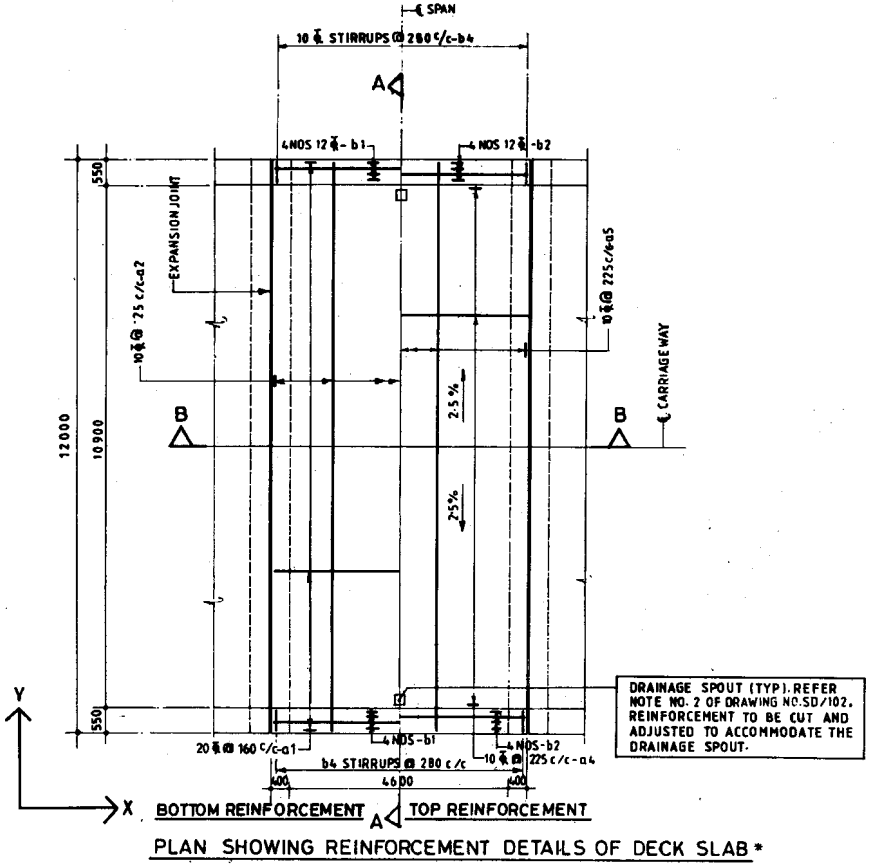


PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO	LOCATION	BAR MKD.	SHAPE	DIA (mm.)	SPACING (mm.)	LENGTH (mm.)	NOS.	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	100, R=80, 5300, 300	20	160	5780	76	439·28	1085
2	SLAB-BOTTOM ALONG-Y	a2	250, R=40, 11900, 250	10	125	12340	44	542·96	335
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23·80	15
4	SLAB-TOP ALONG-X	a4	5200, 300, R=40, 300	10	225	5840	50	292·00	180
5	SLAB-TOP ALONG-Y	a5	5800, 5800, 140	10	225	11600	25	290·00	179
6	KERB - BOTTOM	b1	400, R=50, 5285, 400	12	—	6010	8	48·08	43
7	KERB - TOP	b2	5300, 400, R=50, 400	12	—	6030	8	48·24	43
8	KERB - SIDES	b3	225, R=40, 5268, 225	10	—	5850	8	46·80	29
9	KERB - STIRRUPS	b4	75, 300, 440, 675, 420	10	280 ●	2185	40 ●	87·40	54

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS-REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3 kN/m and wearing coat @ 2 kN/sq.m. = 932 kN

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	5.0
CONCRETE (cu.m.)	32·57
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	2061
ASPHALTIC WEARING COAT (sq.m)	58·86

NOTES :-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note-no(B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502

* S.Reinforcement of adjacent span superstructure, approach slab not shown.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS. (WITHOUT FOOTPATHS)

MKD	DATE	DESCRIPTION	BY

REVISION

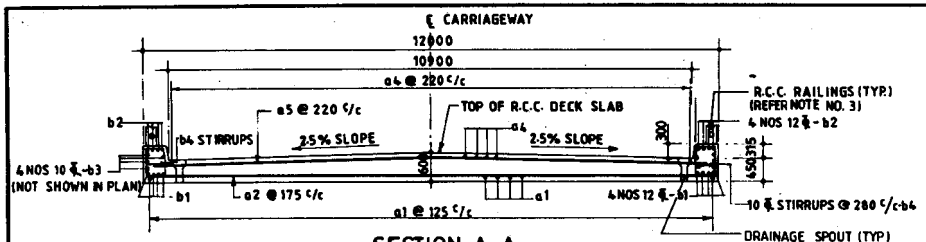
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 5.0m
(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990.
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/109

DRG. NO. SD/110



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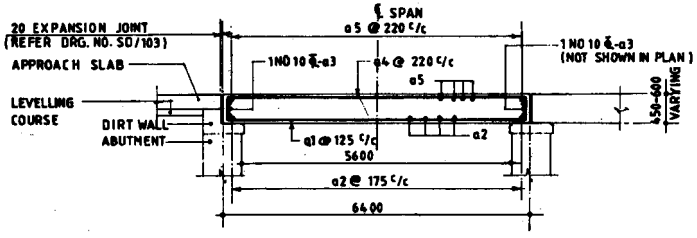

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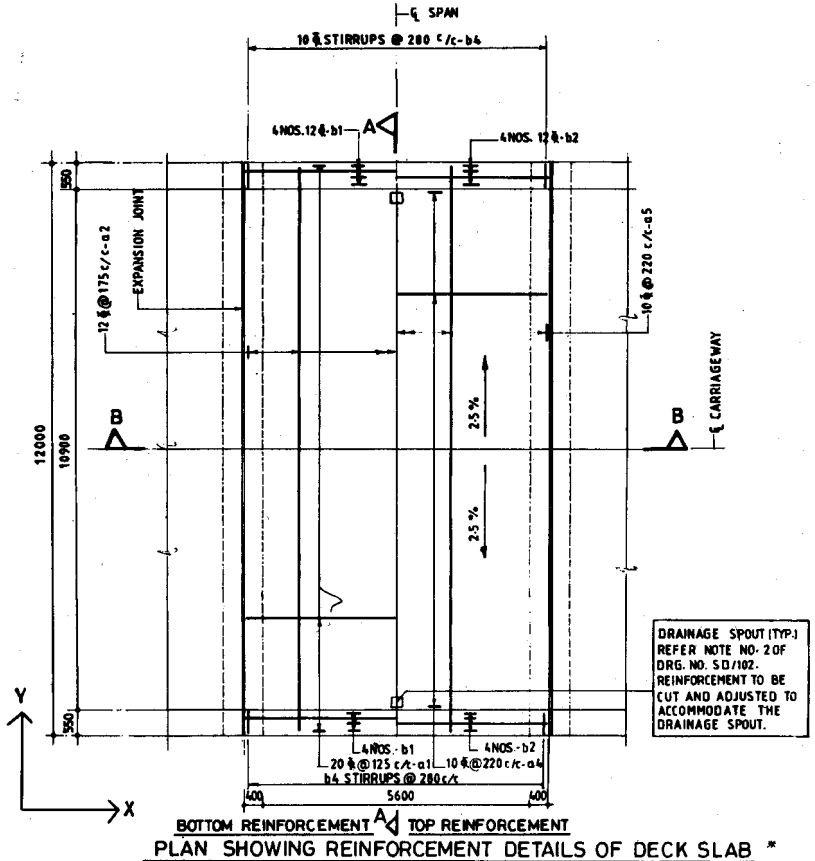


BOTTOM REINFORCEMENT A TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS	TOTAL LENGTH(m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG - X	a1	R=80, 325, 6300, 325	20	125	6830	97	662·51	1636
2	SLAB-BOTTOM ALONG - Y	a2	R=50, 250, 11900, 250	12	175	12330	37	456·21	405
3	SLAB-SIDE ALONG - Y	a3	11900	10	—	11900	2	23·80	15
4	SLAB-TOP ALONG - X	a4	6300, R=40, 325, 325	10	220	6890	51	351·39	217
5	SLAB-TOP ALONG - Y	a5	5800, 5800, 245	10	220	11600	30	348·00	215
6	KERB-BOTTOM	b1	R=50, 400, 6284, 400	12	—	7010	8	56·08	50
7	KERB - TOP	b2	6300, R=50, 400, 400	12	—	7030	8	56·24	50
8	KERB-SIDES	b3	R=40, 225, 6260, 225	10	—	6850	8	54·80	34
9	KERB-STIRRUPS	b4	250, 244, 623, 428	10	280●	2280	48●	109·44	68

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS - REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3kN/m and wearing coat @ 2kN/sq.m. = 1196kN.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN(m)	6·0
CONCRETE (cu.m)	42·43
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	2825
ASPHALTIC WEARING COAT (sq.m)	69·76

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no. (B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per I.S. 2502.

* 5. Reinforcement of adjacent span superstructure, approach slab not shown.

MKD.	DATE	DESCRIPTION	BY

REVISION

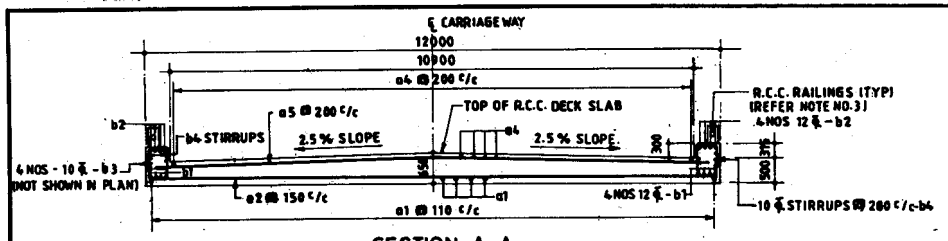
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 6·0m
(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KASTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/110

DRG. NO. SD/111



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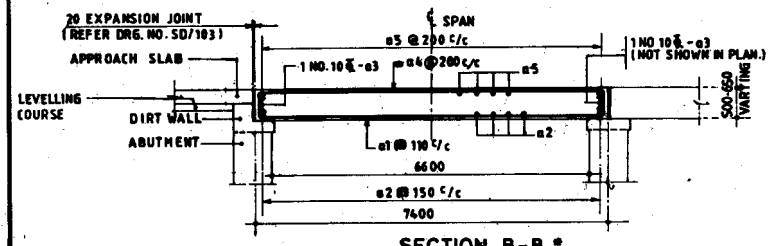

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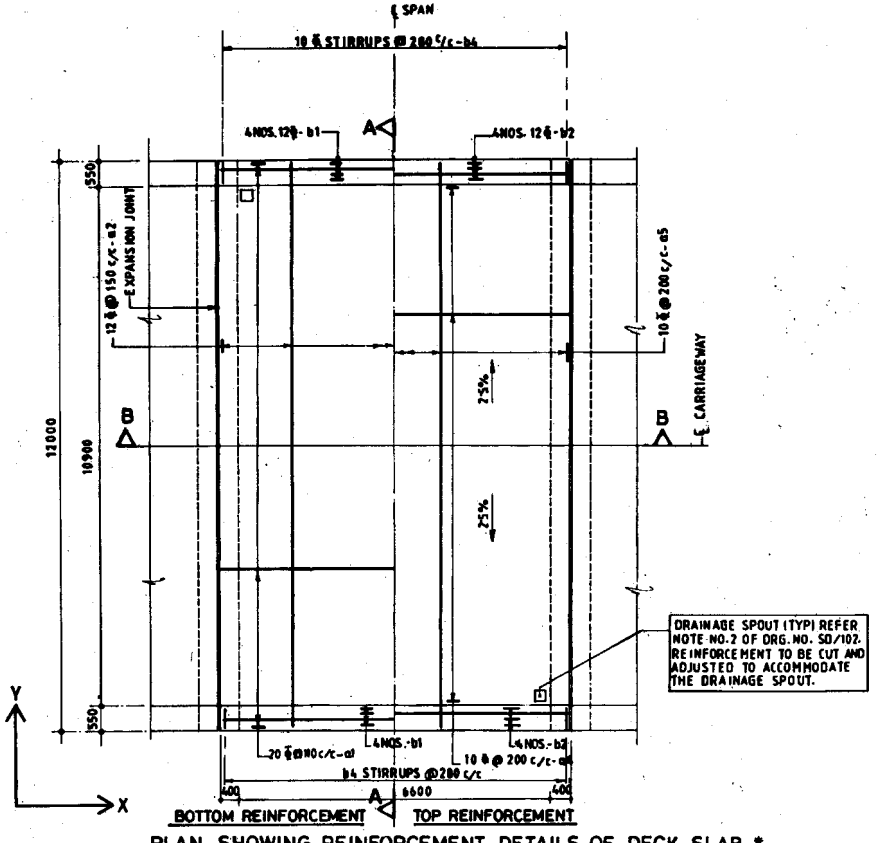


PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	R=80, 7300, 350, 350	20	110	7080	110	866.80	2141
2	SLAB-BOTTOM ALONG-Y	a2	R=50, 11900, 250, 250	12	150	12330	50	616.50	547
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	7300, R=40, 350, 350	10	200	7940	56	444.64	274
5	SLAB-TOP ALONG-Y	a5	5800, 5800, 345	10	200	11600	38	440.80	272
6	KERB-BOTTOM	b1	R=50, 7284, 450, 450	12	—	8110	8	64.88	58
7	KERB-TOP	b2	7300, R=50, 450, 450	12	—	8130	8	65.04	58
8	KERB-SIDES	b3	R=40, 7280, 325, 325	10	—	7850	8	62.80	39
9	KERB-STIRRUPS	b4	450, 450, 250, 250, 125, 125, 450	10	200●	2300	54●	120.52	79

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS - REFER DRG. NO. SD/103.

Dead load of the superstructure per span including R.C.C. railings @ 3kN/m and wearing coat @ 2 kN/sq.m. = 1490 kN

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES.
SD/102	GENERAL ARRANGEMENT.
SD/103 & SD/104	MISCELLANEOUS DETAILS.
SD/105	DETAILS OF R.C.C. RAILINGS. (WITHOUT FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	7.0
CONCRETE (cu.m.)	53.51
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	3657
ASPHALTIC WEARING COAT (sq.m)	80.66

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no (B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
5. Reinforcement of adjacent span superstructure, approach slab not shown.

MKD	DATE	DESCRIPTION	BY

REVISION

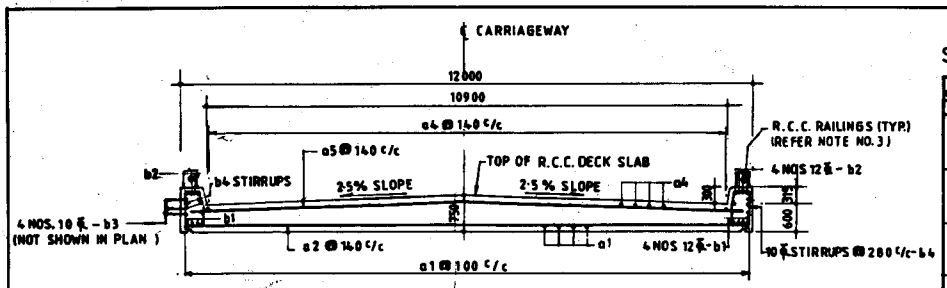
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 7.0m
(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/111

DRG. NO. SD/112



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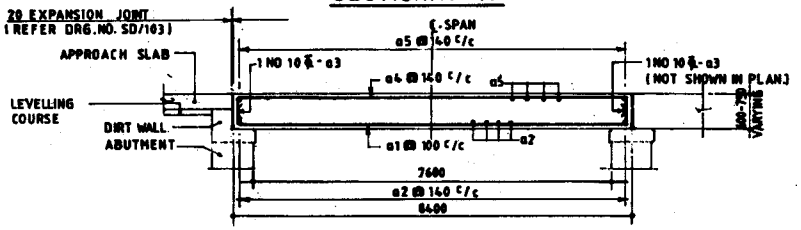

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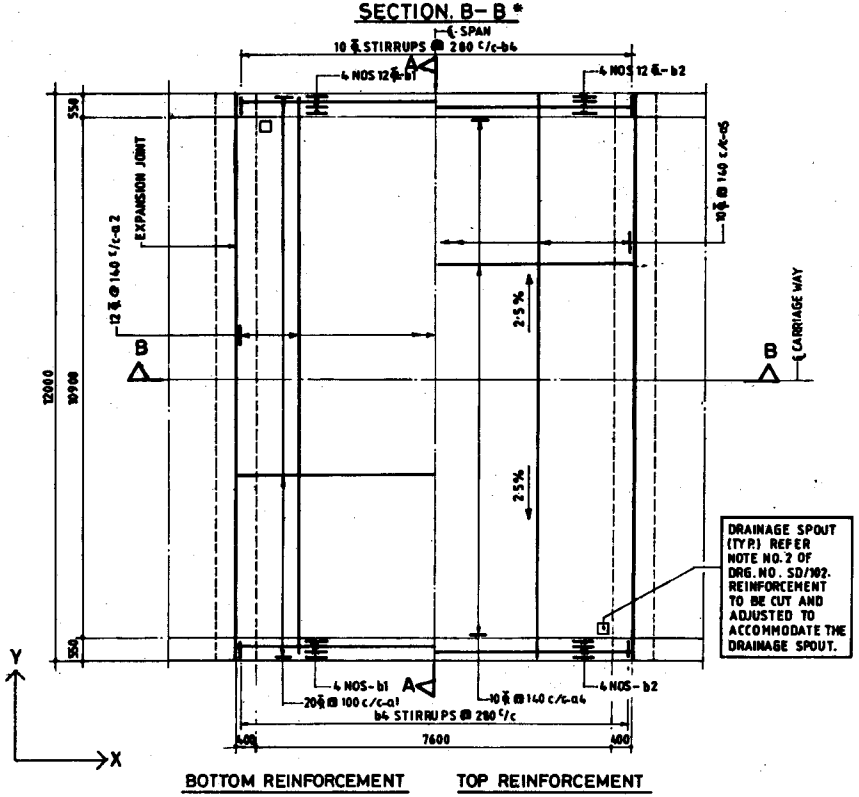


BOTTOM REINFORCEMENT TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL NO	LOCATION	BAR MKD	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	400 R=80 400; 8200	20	100	8980	120	1077.60	2662
2	SLAB-BOTTOM ALONG-Y	a2	300 R=50 300; 11900	12	140	12430	61	758.23	673
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	8300; 400 R=40 400	10	140	9040	79	714.36	441
5	SLAB-TOP ALONG-Y	a5	5800 5800; 745	10	140	11600	61	707.60	437
6	KERB-BOTTOM	b1	300 R=50 300; 8284	12	—	9210	8	73.68	65
7	KERB-TOP	b2	8300; 300 R=50 300	12	—	9230	8	73.84	66
8	KERB-SIDE	b3	215 R=40 215; 8260	10	—	8850	8	70.80	44
9	KERB-STIRRUPS	b4	400 773 450	10	280 ●	2580	62 ●	159.96	99

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS — REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3KN/m and wearing coat @ 2kN/sq m = 1933 kN

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	8·0
CONCRETE (cu. m.)	70·81
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	4727
ASPHALTIC WEARING COAT (sq.m)	91·56

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note. no. (B)-I of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcements of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS.2502.

* 5. Reinforcement of adjacent span superstructure, approach slab not shown.

MKD	DATE	DESCRIPTION	BY

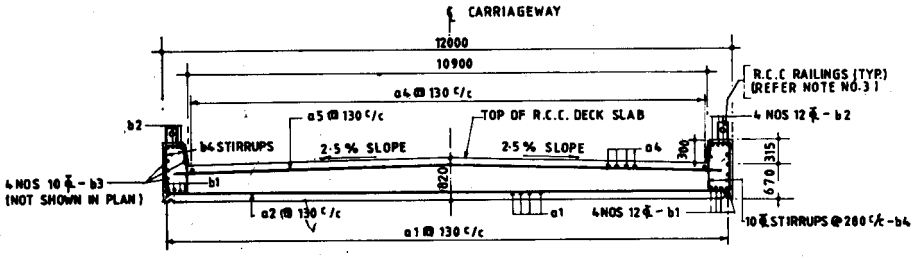
REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPAN 8·0m
(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG NO. SD/112



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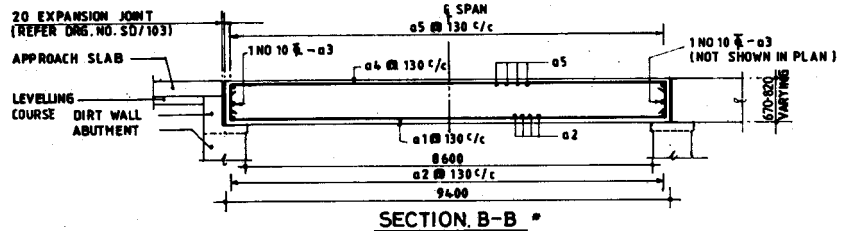

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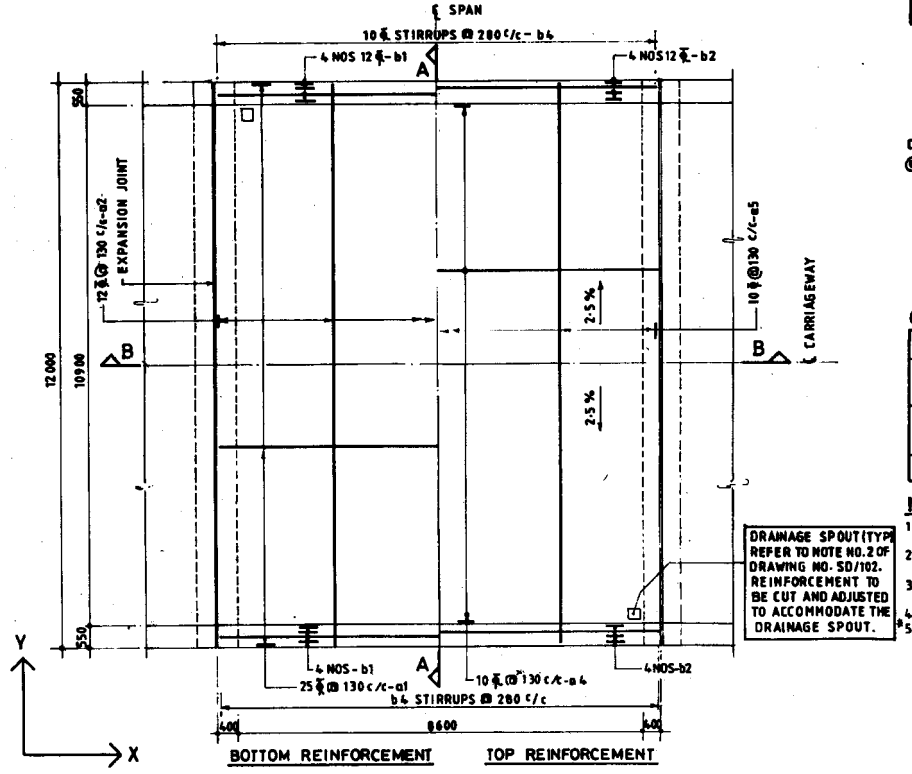


PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH(m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG - X	a1	450, R=100, 9300, 450	25	130	10050	93	934.65	35 98
2	SLAB-BOTTOM ALONG-Y	a2	300, R=50, 11900, 300	12	130	12430	73	907.39	8 06
3	SLAB- SIDE ALONG-Y	a 3	11900	10	—	11900	2	23·80	15
4	SLAB-TOP ALONG-X	a 4	9300, 450, R=40, 450	10	130	10140	85	861.90	5 32
5	SLAB-TOP ALONG-Y	a 5	5800, 5800, 145	10	130	11600	73	846.80	5 22
6	KERB-BOTTOM	b1	600, R=50, 9274, 600	12	—	10400	8	83.20	74
7	KERB-TOP	b2	9300, 600, R=50, 600	12	—	10430	8	83.44	74
8	KERB-SIDES	b3	315, R=40, 9250, 315	10	—	9840	8	78.72	49
9	KERB-STIRRUPS	b4	75, 400, 300, 100, 430	10	280●	2710	70●	189.70	1 17

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS - REFER DRG. NO. SD/103.

Dead load of the superstructure per span including R.C.C. railing @ 3 kN/m and wearing coat @ 2 kN/sq.m. = 2353 kN.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS (WITHOUT FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN(m)	9·0
CONCRETE (cu.m)	87·14
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	6076
ASPHALTIC WEARING COAT (sq.m)	102·46

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note.no.(B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcements of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
5. * Reinforcement of adjacent span superstructure, approach slab not shown.

MKD	DATE	DESCRIPTION	BY

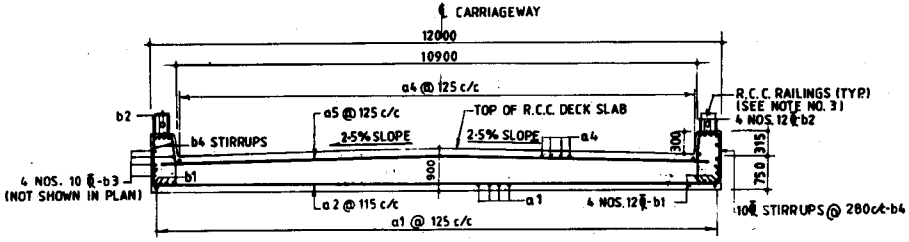
REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

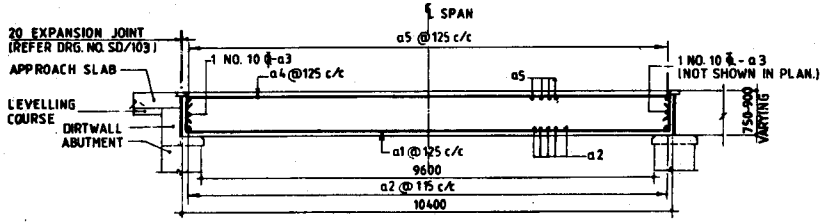
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 9·0m
(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/113



SECTION A-A



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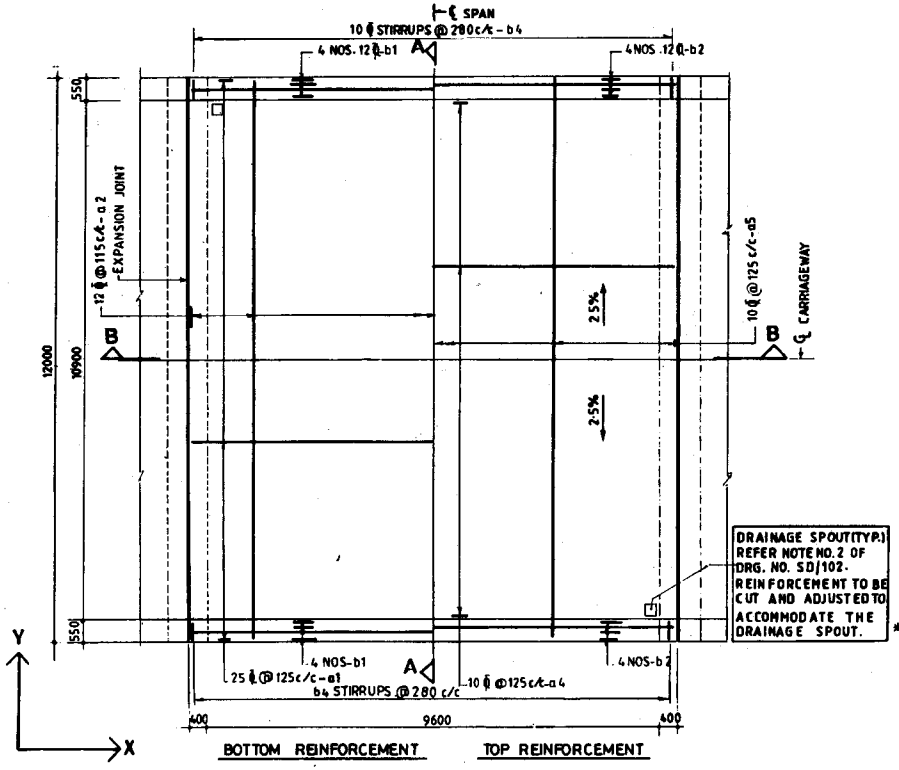


BOTTOM REINFORCEMENT TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO	LOCATION	BAR MKD	SHAPE	DIA. (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH(m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	500, R=100, 10300, 500	25	125	11150	96	1070.40	4121
2	SLAB-BOTTOM ALONG-Y	a2	300, R=80, 11900, 300	12	115	12430	91	1131.13	1004
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	10300, 500, R=40, 500	10	125	11240	89	1000.36	617
5	SLAB-TOP ALONG-Y	a5	5800, 5800, 142	10	125	11600	84	974.40	601
6	KERB-BOTTOM	b1	650, R=50, 10275, 650	12	–	11500	8	92.00	82
7	KERB-TOP	b2	10300, 650, R=50, 650	12	–	11530	8	92.24	82
8	KERB-SIDES	b3	325, R=40, 10250, 325	10	–	10840	8	86.72	54
9	KERB-STIRRUPS	b4	274, 203, 643, 318, 536	10	280●	2870	76●	218.12	135

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS-REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3 kN/m. and wearing coat @ 2 kN/sq m = 2843 kN.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/105	DETAILS OF R.C.C. RAILINGS. (WITHOUT FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	10.0
CONCRETE (cu. m.)	106.40
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	7047
ASPHALTIC WEARING COAT (sq. m.)	113.36

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note, no.(B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcements of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/105 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
5. * Reinforcement of adjacent span superstructure, approach slab not shown.

MKD.	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

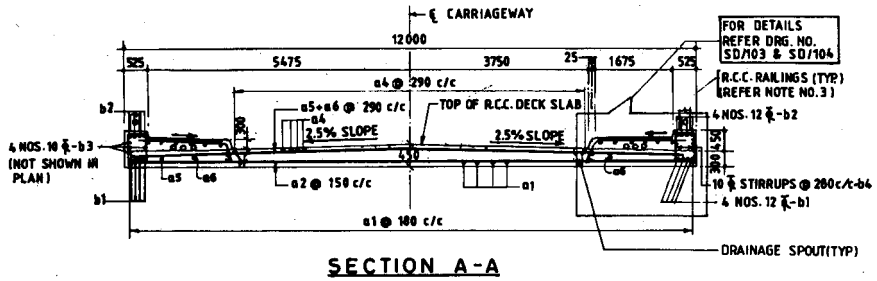
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 10·0m

(WITHOUT FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(S.S. REDDY) E.E.	(S.K. KAISTHA) S.E.	(P.N. SHIVAPRASAD) C.E.	DRG. NO SD/114

DRG. NO. SD/115



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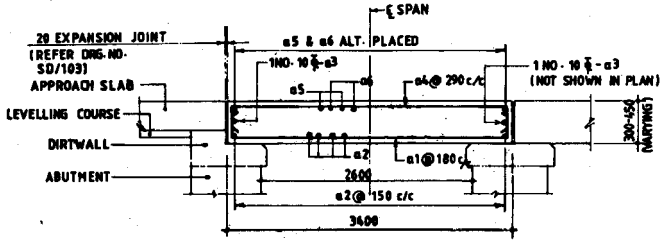

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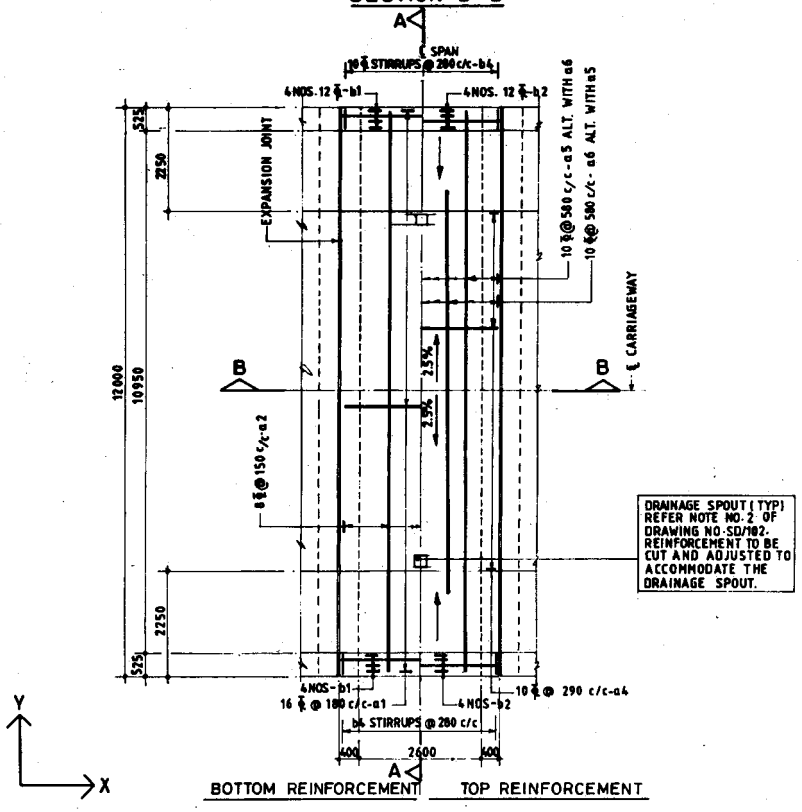


BOTTOM REINFORCEMENT TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH(m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	200, R=80, 200; 3300	16	180	3600	67	241.20	381
2	SLAB-BOTTOM ALONG-Y	a2	200, R=30, 200; 11900	8	150	12250	23	281.75	111
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	200, R=40, 200; 3300	10	290	3640	27	98.28	61
5	SLAB-TOP ALONG-Y	a5	5800, 5800; 250	10	580	11600	7	81.20	50
6	SLAB-TOP ALONG-Y	a6	4050, 4050; 101	10	580	8100	7	56.70	35
7	KERB-BOTTOM	b1	350, R=50, 350; 3292	12	—	3920	8	31.36	28
8	KERB-TOP	b2	350, R=50, 350; 3300	12	—	3930	8	31.44	28
9	KERB-SIDES	b3	300, R=40, 300; 3268	10	—	3810	8	30.48	19
10	KERB-STIRRUPS	b4	40, 616	10	280 ●	2240	26 ●	58.24	36
11	FOOTPATH-TOP ALONG-Y	f1	542, 2050, R=30, 300	8	200	2840	36	102.24	40
12	FOOTPATH-TOP ALONG-X	f2	250, R=30, 250; 3300	8	—	3750	20	75.00	30
13	FOOTPATH-MIDDLE ALONG-X	f3	3268	10	—	3268	16	52.29	32
14	FOOTPATH-EDGES ALONG-X	f4	300, R=30, 1500; 3268	8	—	4220	4	16.88	7

● DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS- REFER DRG. NO. SD/103.

Dead load of the superstructure per span including R.C.C. railings @ 3kN/m and wearing coat @ 2kN/sq.m = 571 kN

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	3.0
CONCRETE (cu.m.)	20.44
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg.)	917
ASPHALTIC WEARING COAT (sq.m.)	25.50

NOTES:—

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no. (B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
5. * Reinforcement of adjacent span superstructure, approach slab not shown.
6. * Service ducts and reinforcement of footpath not shown in plan.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES.
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS.
SD/106	DETAILS OF R.C.C. RAILINGS (WITH FOOTPATHS)

MKD.	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

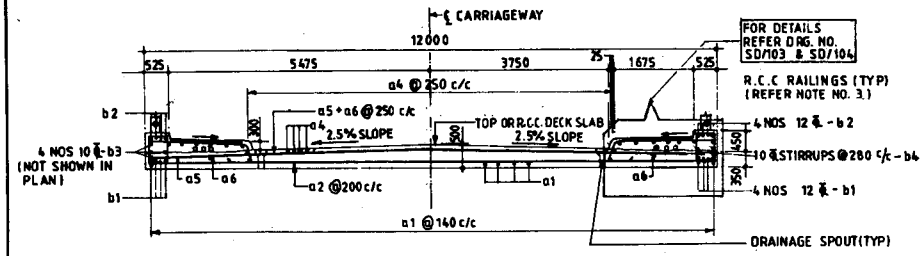
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPAN 3.0m

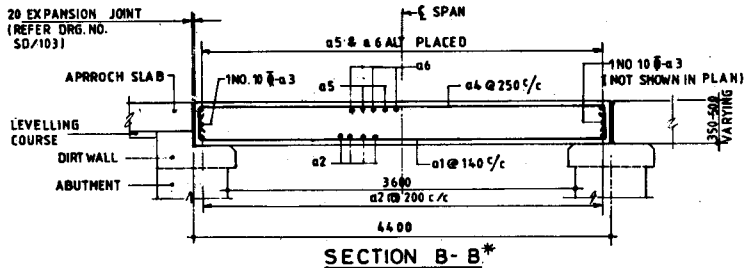
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/115

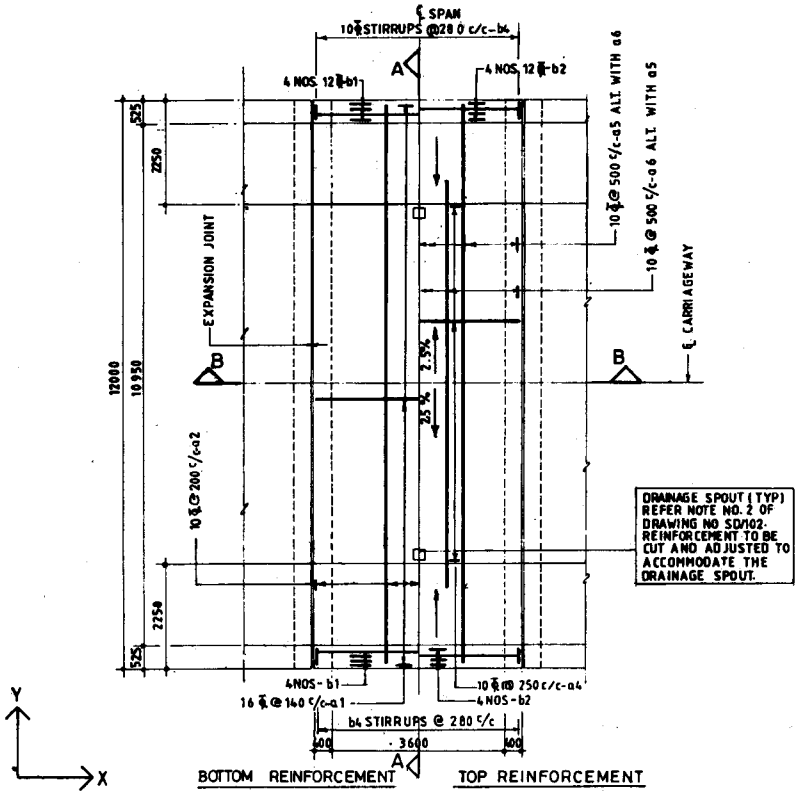
DRG. NO. SD/116



SECTION A-A



SECTION B-B*



PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB*

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH(m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	250 R=60 4300 250	16	140	4700	86	404·2	639
2	SLAB-BOTTOM ALONG-Y	a2	200 R=40 11900 200	10	200	12240	23	281·52	174
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23·80	15
4	SLAB-TOP ALONG-X	a4	4300 250 R=40 250	10	250	4740	31	146·94	91
5	SLAB-TOP ALONG-Y	a5	5800 5800 45	10	500	11600	10	116·000	72
6	SLAB-TOP ALONG-Y	a6	4050 4050 101	10	500	8100	10	81·00	50
7	KERB-BOTTOM	b1	350 R=50 4292 350	12	—	4920	8	39·36	35
8	KERB-TOP	b2	4300 350 R=50 350	12	—	4930	8	39·44	35
9	KERB-SIDES	b3	325 R=40 4268 325	10	—	4860	8	38·88	24
10	KERB-STIRRUPS	b4	405 340 486	10	280 ⊛	2340	34 ⊛	79·56	49
11	FOOTPATH-TOP ALONG-Y	f1	54 2050 R=30 300	8	200	2895	46	133·17	53
12	FOOTPATH-TOP ALONG-X	f2	4300 250 R=30 250	8	—	4750	20	95·00	38
13	FOOTPATH-MIDDLE ALONG-X	f3	4268	10	—	4268	16	68·29	42
14	FOOTPATH-EDGES ALONG-X	f4	500 R=30 4268 500	8	—	5220	4	20·88	8

⊛ DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS - REFER DRG NO. SD/103.

Dead load of the superstructure per span including R.C.C. railings @ 3kN/m and wearing coat @ 2kN/sq.m. = 791 kN

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/106	DETAILS OF R.C.C RAILINGS (WITH FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	4.0
CONCRETE (cu.m.)	29·09
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	1391
ASPHALTIC WEARING COAT (sq.m.)	33·00

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no.(B)1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
*5. Reinforcement of adjacent span superstructure, approach slab not shown.
*6. Service ducts and reinforcement of footpath not shown in plan.

MKD.	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

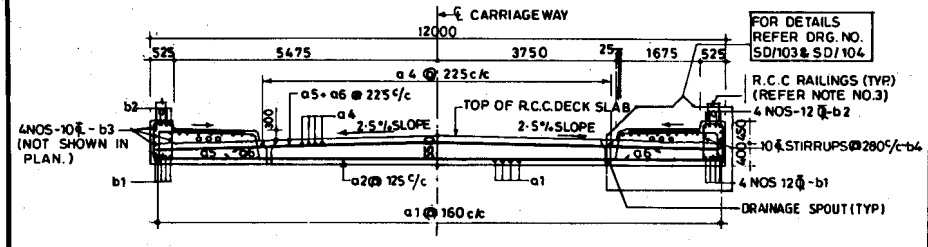
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPAN 4.0m

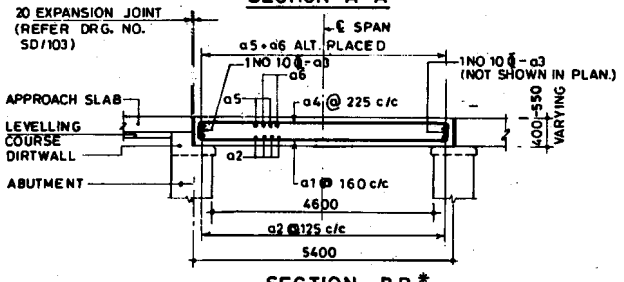
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/116

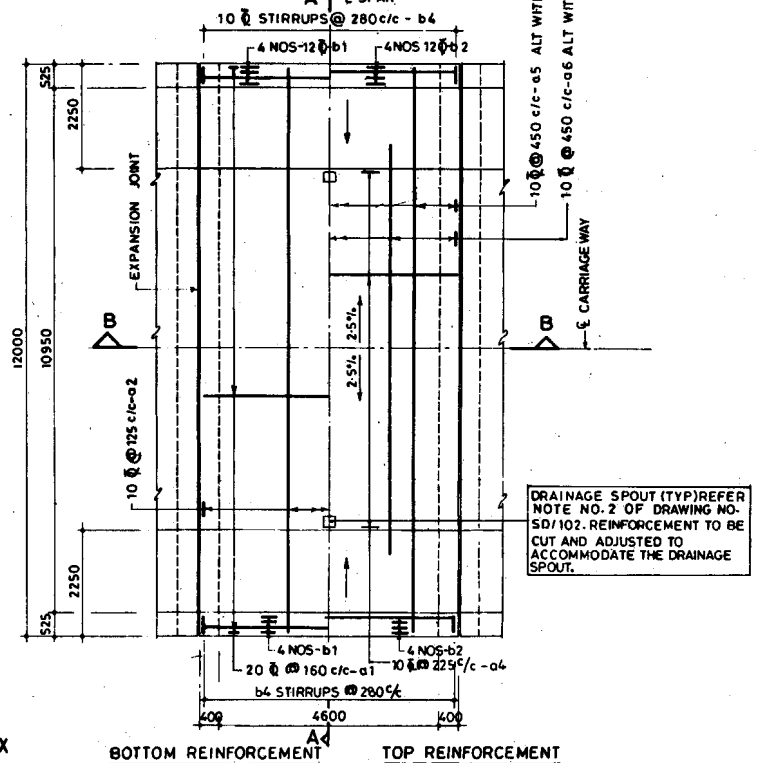
DRG. NO. SD/117



SECTION A-A



SECTION B-B*



BOTTOM REINFORCEMENT TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB*

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH(m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	300 R-80 5300 300	20	160	5780	76	439.28	1085
2	SLAB-BOTTOM ALONG-Y	a2	250 R-40 11900 250	10	125	12340	44	542.96	335
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	300 5300 R-40 300	10	225	5840	35	204.40	126
5	SLAB-TOP ALONG-Y	a5	5800 5800	10	450	11600	13	150.80	93
6	SLAB-TOP ALONG-Y	a6	4050 4050	10	450	8100	13	105.30	65
7	KERB-BOTTOM	b1	300 R=50 5285 300	12	—	6010	8	48.08	43
8	KERB-TOP	b2	400 5300 R=50 400	12	—	6030	8	48.24	43
9	KERB-SIDES	b3	325 R-40 5260 325	10	—	5850	8	46.80	29
10	KERB-STIRRUPS	b4	415 710	10	280 ⊛	2430	40 ⊛	97.20	60
11	FOOTPATH-TOP ALONG-Y	f1	639 2050 R-30 300 10	8	200	2940	56	164.64	65
12	FOOTPATH-TOP ALONG-X	f2	300 5300 R-30 300	8	—	5850	20	117.00	46
13	FOOTPATH-MIDDLE ALONG-X	f3	5260	10	—	5260	16	84.16	52
14	FOOTPATH-EDGES ALONG-X	f4	500 R-30 5260 300	8	—	6210	4	24.84	10

⊛ DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS – REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3kN/m and wearing coat @ 2kN/sq.m. = 1062 kN

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	5·0
CONCRETE (cu.m)	38·95
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	2170
ASPHALTIC WEARING COAT (Sq.m)	40·50

NOTES:–

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no.⊛-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
5. * Reinforcement of adjacent span superstructure, approach slab not shown.
6. * Service ducts and reinforcement of footpath not shown in plan.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES.
SD/102	GENERAL ARRANGEMENT.
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/106	DETAILS OF R.C.C. RAILINGS (WITH FOOTPATHS)

MKD	DATE	DESCRIPTION	BY

REVISION

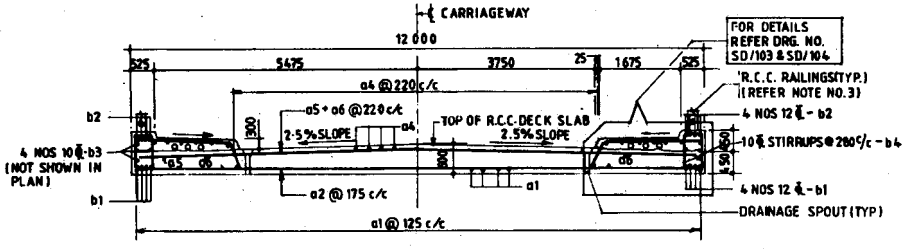
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

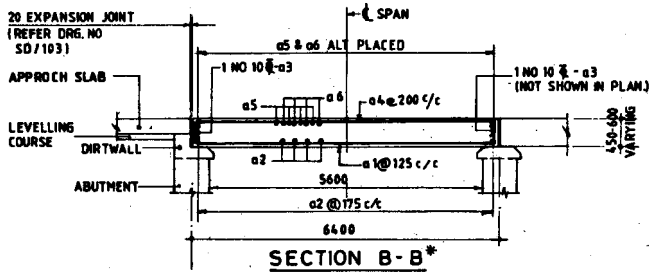
R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 5·0m
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. RAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/117

DRG. NO. SD/118



SECTION A-A



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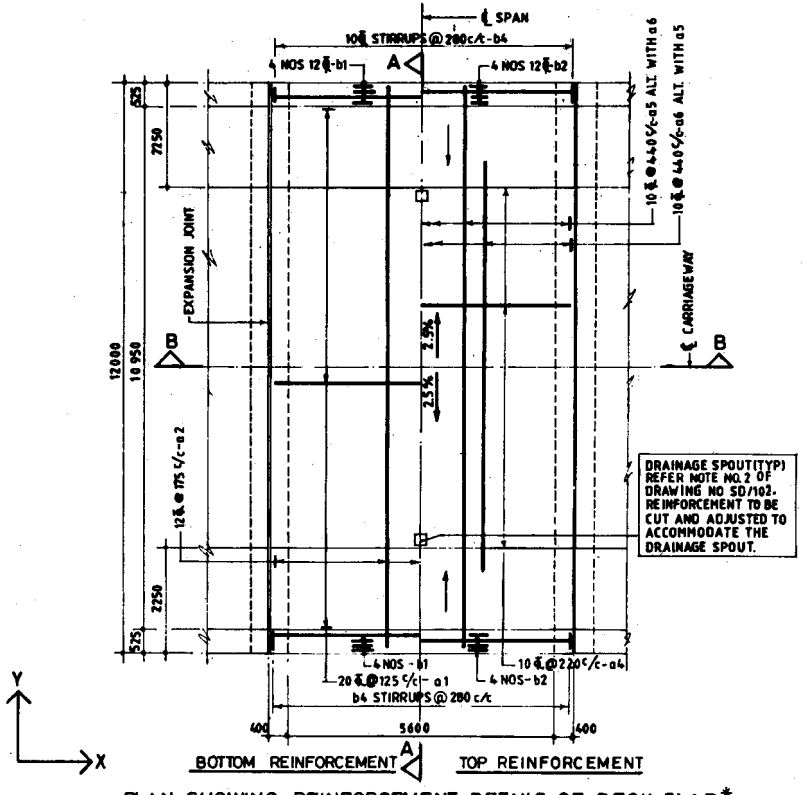


PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB*

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION	BAR MKD.	SHAPE	DIA. (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH(m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	325, R=80, 6300, 325	20	125	6830	97	662·51	1636
2	SLAB-BOTTOM ALONG-Y	a2	250, R=50, 11900, 250	12	175	12330	37	456·21	405
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23·80	15
4	SLAB-TOP ALONG-X	a4	6300, 325, R=40, 325	10	220	6890	35	241·15	149
5	SLAB-TOP ALONG-Y	a5	5800, 5800, 145	10	440	11600	16	185·60	115
6	SLAB-TOP ALONG-Y	a6	4050, 4050, 100	10	440	8100	16	129·60	80
7	KERB-BOTTOM	b1	400, R=50, 6284, 400	12	—	7010	8	56·08	50
8	KERB-TOP	b2	400, 6300, R=80, 400	12	—	7030	8	56·24	50
9	KERB-SIDES	b3	325, R=40, 6260, 325	10	—	6850	8	54·80	34
10	KERB-STIRRUPS	b4	403, 175	10	280 ⊛	2520	48 ⊛	120·96	75
11	FOOTPATH-TOP ALONG-Y	f1	82, 1000, R=30, 800	8	200	2990	66	197·34	78
12	FOOTPATH-TOP ALONG-X	f2	300, 6300, R=30, 300	8	—	6850	20	137·00	54
13	FOOTPATH-MIDDLE ALONG-X	f3	6260	10	—	6260	16	100·16	62
14	FOOTPATH-EDGES ALONG-X	f4	500, R=30, 6260, 500	8	—	7210	4	28·84	11

⊛ DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS – REFER DRG. NO. SD/103.

Dead Load of the superstructure per span including R.C.C railings @ 3 kN/m and wearing coat @ 2 kN/sq.m. = 1334 kN

QUANTITIES (PER SPAN)

EFFECTIVE SPAN(m)	6·0
CONCRETE (cu. m.)	50·00
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	2955
ASPHALTIC WEARING COAT (sq.m.)	48·00

NOTES:–

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no (B) 1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per I S 2502
5. * Reinforcement of adjacent span superstructure, approach slab not shown.
6. * Service ducts and reinforcement of footpath not shown in plan.

REFERENCE DRAWINGS

DRAWING NO	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/106.	DETAILS OF R.C.C. RAILINGS (WITH FOOTPATHS)

MKD	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

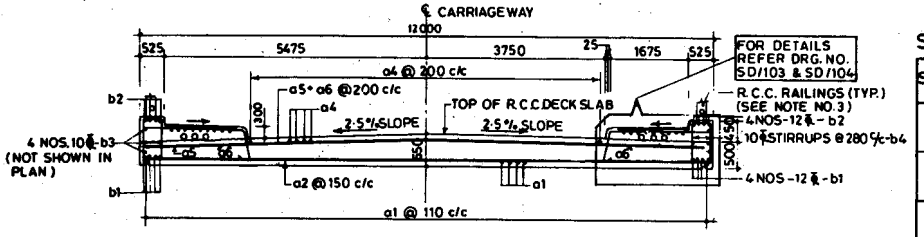
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPAN 6.0m

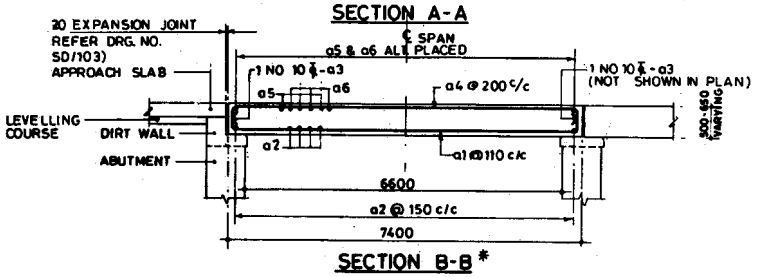
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/118

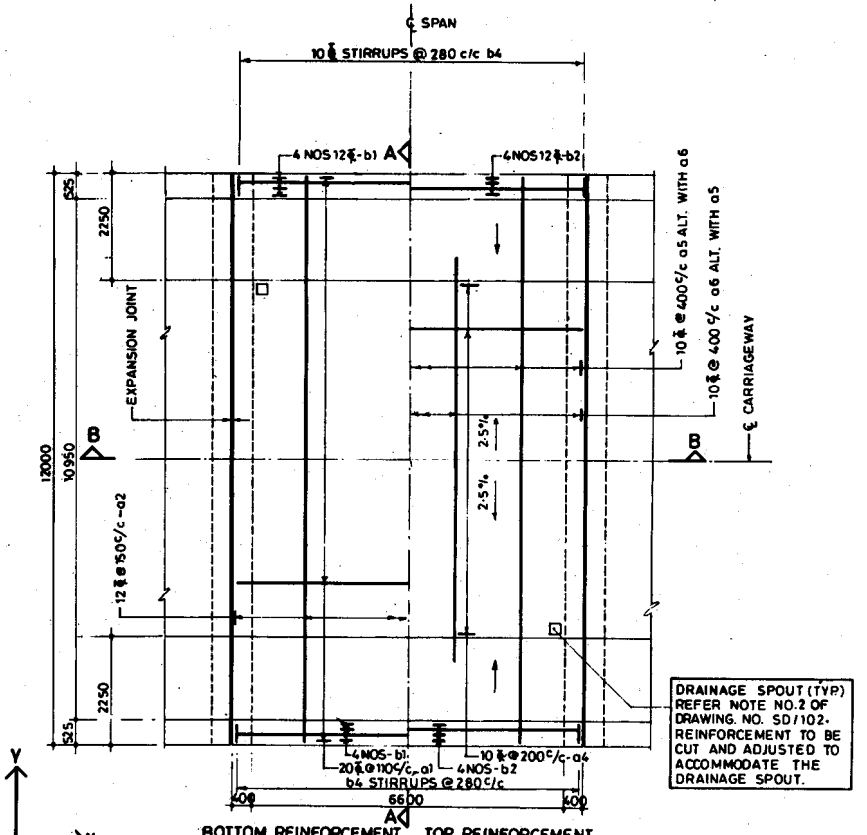
DRG. NO. SD/119



SECTION A-A



SECTION B-B *



BOTTOM REINFORCEMENT TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB*

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION	BAR MKD.	SHAPE	DIA. (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	350 R=80 350 / 7300	20	110	7880	110	866.80	2141
2	SLAB-BOTTOM ALONG-Y	a2	250 R=50 250 / 11900	12	150	12330	50	616.50	547
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	7300 / 350 R=40 350	10	200	7940	39	309.66	191
5	SLAB-TOP ALONG-Y	a5	5800 5800 / 250	10	400	11600	20	232.00	143
6	SLAB-TOP ALONG-Y	a6	4050 4050 / 150	10	400	8100	20	162.00	100
7	KERB-BOTTOM	b1	450 R=50 450 / 7284	12	—	8110	8	64.88	58
8	KERB-TOP	b2	7300 / 450 R=50 450	12	—	8130	8	65.04	58
9	KERB-SIDES	b3	325 R=40 325 / 7260	10	—	7850	8	62.80	39
10	KERB-STIRRUPS	b4	403 R=40 808	10	280 ⊛	2620	54 ⊛	141.48	87
11	FOOTPATH-TOP ALONG Y	f1	65 2050 R=30 279 200	8	200	3021	76	229.60	91
12	FOOTPATH-TOP ALONG X	f2	7300 / 350 R=30 350	8	—	7950	20	159.00	63
13	FOOTPATH-MIDDLE ALONG-X	f3	7260	10	—	7260	16	116.16	72
14	FOOTPATH-EDGES ALONG-X	f4	500 R=80 500 / 7260	8	—	8210	4	32.84	13

⊛ DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS — REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C railings @ 3 kN/m and wearing coat @ 2kN/sq.m. = 1668 kN.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/106	DETAILS OF R.C.C RAILINGS (WITH FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	7·0
CONCRETE (cu.m.)	62·25
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	3794
ASPHALTIC WEARING COAT (sq.m.)	55·50

NOTES :-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no. (B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per I S 2502.

*5. Reinforcement of adjacent span superstructure, approach slab not shown.

*6. Service ducts and reinforcement of footpath not shown in plan.

MKD	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

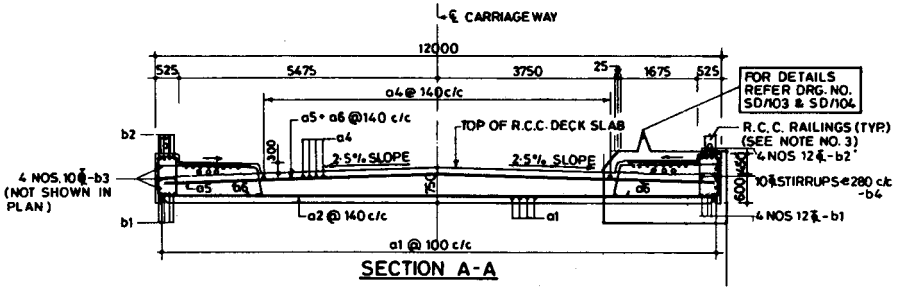
STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 7·0m

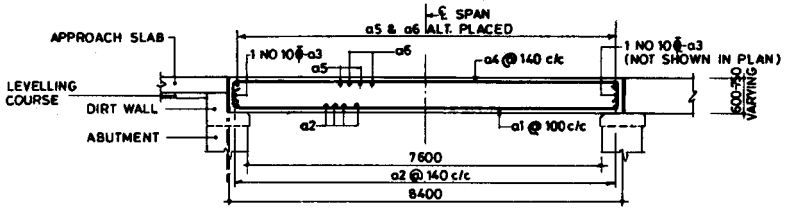
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKUMAR) E.E	(S. K. KAISTHA) S.E.	(M. K. MUKHERJEE) C.E.	DRG. NO. SD/119

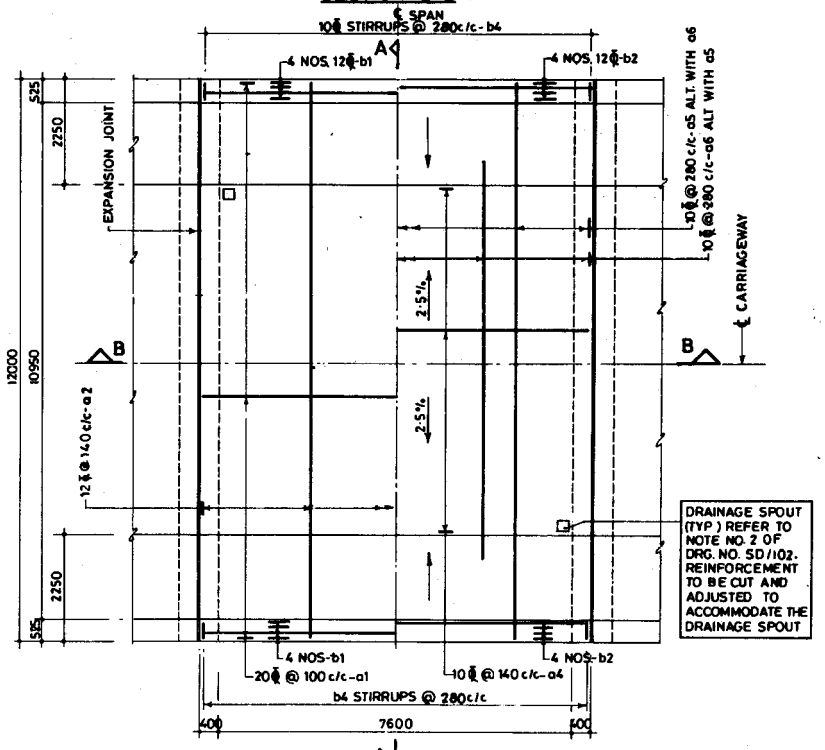
DRG. NO. SD/120



SECTION A-A



SECTION B-B *



PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION.	BAR MKD.	SHAPE	DIA (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	400 R = 80 400 / 8300	20	100	8980	120	1077.60	2662
2	SLAB-BOTTOM ALONG-Y	a2	300 R=50 300 / 11900	12	140	12430	61	758.23	673
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	8300 / 400 R=40 400	10	140	9040	55	497.20	307
5	SLAB-TOP ALONG-Y	a5	5800 5800	10	280	11600	31	359.60	222
6	SLAB-TOP ALONG-Y	a6	4050 4050	10	280	8100	31	251.10	155
7	KERB-BOTTOM	b1	500 R=50 500 / 8284	12	—	9210	8	73.70	65
8	KERB-TOP	b2	8300 / 500 R=50 500	12	—	9230	8	73.84	66
9	KERB-SIDES	b3	325 R=40 325 / 8260	10	—	8850	8	70.80	44
10	KERB-STIRRUPS	b4	403 / 340 / 90	10	280 ⊛	2820	62 ⊛	174.84	108
11	FOOTPATH-TOP ALONG-Y	f1	78 / 20 / 50 / 30 / 300 / 90	8	200	3140	86	270.04	107
12	FOOTPATH-TOP ALONG-X	f2	8300 / 400 R=30 400	8	—	9050	20	181.00	72
13	FOOTPATH-MIDDLE ALONG-X	f3	8260	10	—	8260	16	132.16	82
15	FOOTPATH-EDGES ALONG-X	f4	500 R=30 500 / 8260	8	—	9210	4	36.84	15

⊛ DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS – REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C railings @ 3 kN/m and wearing coat @ 2 kN/sq.m. = 2114 kN

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	8·0
CONCRETE (cu.m.)	80·74
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	4823
ASPHALTIC WEARING COAT (Sq.m)	63·00

NOTES:-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no.(B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per I S 2502
5. * Reinforcement of adjacent span superstructure, approach slab not shown.
6. * Service ducts and reinforcement of footpath not shown in plan.

REFERENCE DRAWINGS

DRAWING NO	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/106	DETAILS OF R.C.C RAILINGS (WITH FOOTPATHS)

MKD	DATE	DESCRIPTION	BY

REVISION

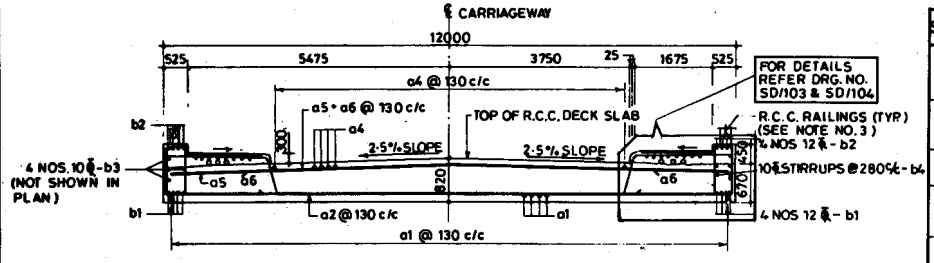
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

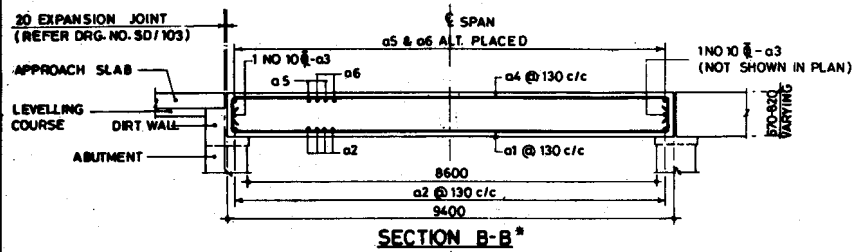
R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 8·0m
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/120

DRG. NO. SD/121



SECTION A-A



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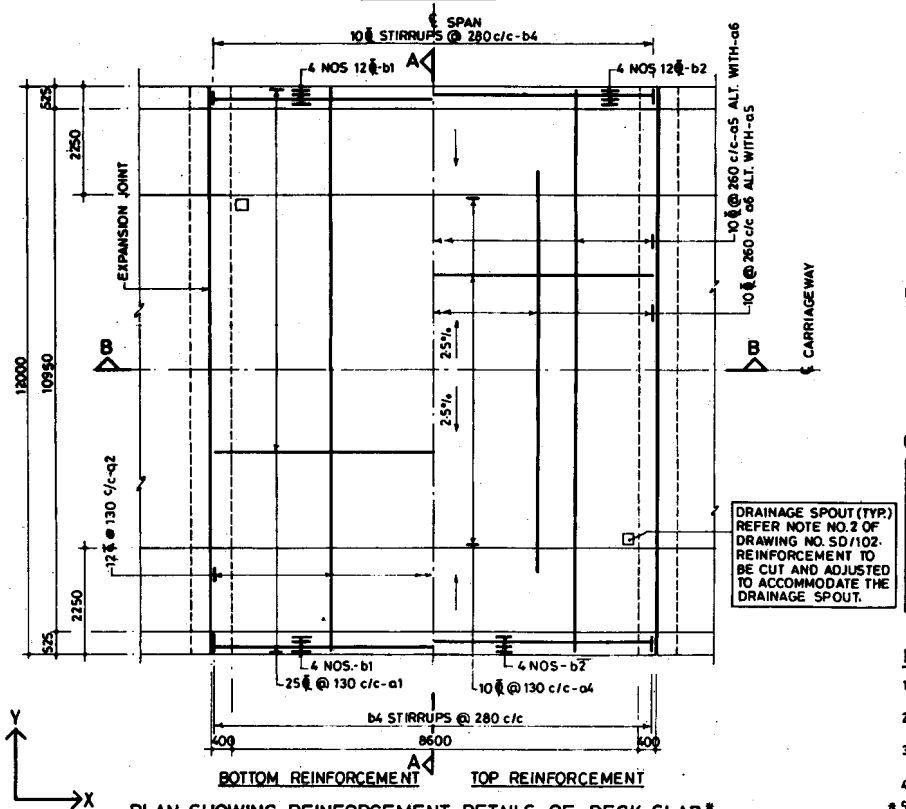


PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB*

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION	BAR MKD	SHAPE	DIA. (mm.)	SPACING (mm.)	LENGTH (mm.)	NOS.	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB BOTTOM ALONG-X	a1	450, R=100, 450; 9300	25	130	10050	93	934.65	3598
2	SLAB-BOTTOM ALONG-Y	a2	300, R=50, 300; 11900	12	130	12430	73	907.39	806
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23.80	15
4	SLAB-TOP ALONG-X	a4	9300; 450, R=40, 450	10	130	10140	59	598.26	369
5	SLAB-TOP ALONG-Y	a5	5800, 5800; 325	10	260	11600	37	429.20	265
6	SLAB-TOP ALONG-Y	a6	4050, 4050; 325	10	260	8100	37	299.70	185
7	KERB - BOTTOM	b1	500, R=50, 500; 9276	12	—	10400	8	83.20	74
8	KERB - TOP	b2	9300; 500, R=50, 500	12	—	10430	8	83.44	74
9	KERB - SIDES	b3	325, R=40, 325; 9250	10	—	9840	8	78.72	49
10	KERB - STIRRUPS	b4	403; 973	10	280 ⊛	2950	70 ⊛	206.50	127
11	FOOTPATH - TOP ALONG-Y	f1	87, 3050, 87, R=30, 300	8	200	3210	96	308.16	122
12	FOOTPATH - TOP ALONG-X	f2	9300; 500, R=30, 500	8	-	10250	20	205.00	81
13	FOOTPATH - MIDDLE ALONG-X	f3	9250	10	-	9250	16	148.00	91
14	FOOTPATH - EDGES ALONG-X	f4	500, R=30, 500; 9250	8	-	10200	4	40.80	16

⊛ DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS - REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C railings @ 3kN/m and wearing coat @ 2 kN/sq.m. = 2580 kN

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS.
SD/106	DETAILS OF R.C.C RAILINGS (WITH FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m)	9·0
CONCRETE (cu.m.)	98·26
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	6166
ASPHALTIC WEARING COAT (sq.m.)	70·50

NOTES:—

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no(B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502.

*5. Reinforcement of adjacent span superstructure, approach slab not shown.
*6. Service ducts and reinforcement of footpath not shown in plan.

MKD	DATE	DESCRIPTION	BY

REVISION

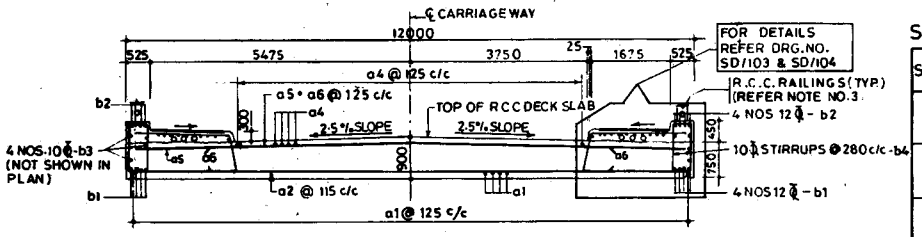
GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

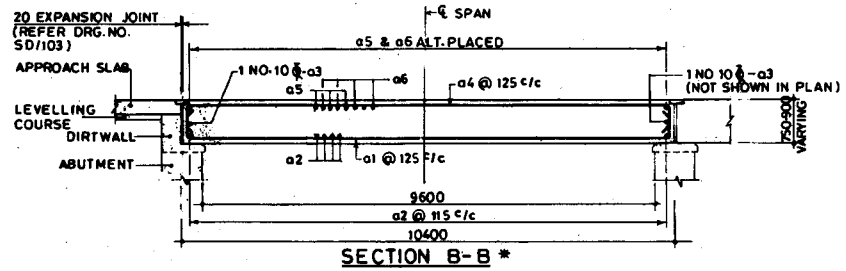
R.C.C. SOLID SLAB SUPERSTRUCTURE (RIGHT) SPAN 9·0m
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(U. JAYAKODI) E.E.	(S.K. KAISTHA) S.E.	(M.K. MUKHERJEE) C.E.	DRG. NO. SD/121

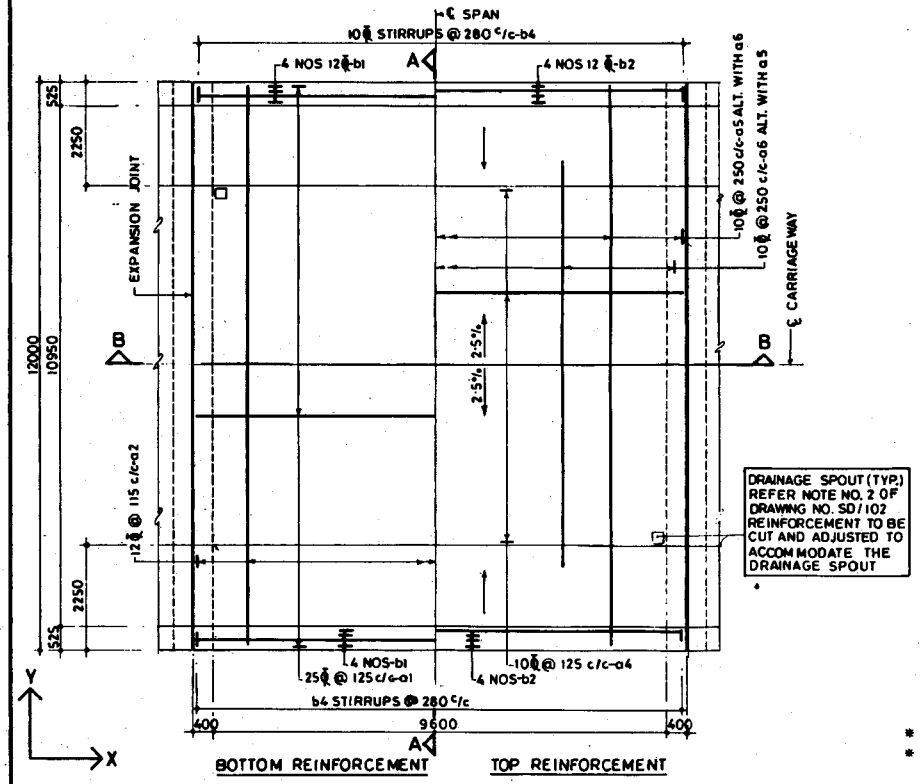
DRG. NO. SD/122



SECTION A-A



SECTION B-B *



BOTTOM REINFORCEMENT TOP REINFORCEMENT

PLAN SHOWING REINFORCEMENT DETAILS OF DECK SLAB *

SCHEDULE OF REINFORCEMENT (PER SPAN)

SL.NO.	LOCATION	BAR MKD	SHAPE	DIA. (mm)	SPACING (mm)	LENGTH (mm)	NOS.	TOTAL LENGTH (m)	WEIGHT (kg)
1	SLAB-BOTTOM ALONG-X	a1	500, R=100, 500; 10300	25	125	11150	96	1070·40	4121
2	SLAB-BOTTOM ALONG-Y	a2	300, R=50, 300; 11900	12	115	12430	91	1131·13	1004
3	SLAB-SIDE ALONG-Y	a3	11900	10	—	11900	2	23·80	15
4	SLAB-TOP ALONG-X	a4	10300; 500, R=40, 500	10	125	11240	61	685·64	423
5	SLAB-TOP ALONG-Y	a5	5800, 5800, 45	10	250	11600	42	487·20	301
6	SLAB-TOP ALONG-Y	a6	4050, 4050, 45	10	250	8100	42	340·20	210
7	KERB-BOTTOM	b1	650, R=50, 650; 10275	12	—	11500	8	92·00	82
8	KERB-TOP	b2	10300; 650, R=50, 650	12	—	11530	8	92·24	82
9	KERB-SIDES	b3	125, R=40, 125; 10250	10	—	10840	8	86·72	54
10	KERB-STIRRUPS	b4	400, 105, R=40	10	280 ⊕	3110	76 ⊕	236·36	146
11	FOOTPATH-TOP ALONG-Y	f1	88, 224, 2050, 300, R=30	8	200	3275	106	347·15	137
12	FOOTPATH-TOP ALONG-X	f2	10300; 600, R=30, 600	8	—	11450	20	229·00	90
13	FOOTPATH-MIDDLE ALONG-X	f3	10250	10	—	10250	16	164·00	101
14	FOOTPATH-EDGES ALONG-X	f4	300, R=30, 300; 10250	8	—	11200	4	44·80	18

⊕ DOES NOT INCLUDE THE ADDITIONAL STIRRUPS AT LAPS OF LONGITUDINAL BARS - REFER DRG. NO. SD/103

Dead load of the superstructure per span including R.C.C. railings @ 3 kN/m and wearing coat @ 2 kN/sq m = 3094. kN.

REFERENCE DRAWINGS

DRAWING NO.	TITLE
SD/101	GENERAL NOTES
SD/102	GENERAL ARRANGEMENT
SD/103 & SD/104	MISCELLANEOUS DETAILS
SD/106	DETAILS OF R.C.C. RAILINGS (WITH FOOTPATHS)

QUANTITIES (PER SPAN)

EFFECTIVE SPAN (m.)	10·0
CONCRETE (cu.m.)	118·70
STEEL INCLUDING 5% EXTRA FOR LAPS AND WASTAGE (kg)	7124
ASPHALTIC WEARING COAT (sq.m.)	78·00

NOTES :-

1. All dimensions are in millimetres unless otherwise mentioned. Only written dimensions are to be followed.
2. Special attention is invited to note no.(B)-1 of drawing no SD/101 regarding the design mix to be adopted.
3. The reinforcement of railing posts shall be incorporated before casting of the deck slab. The railings shall conform to drawing no. SD/106 or any other approved type.
4. Dimensions in schedule of reinforcement are given as per IS 2502
5. * Reinforcement of adjacent span superstructure, approach slab not shown.
6. * Service ducts and reinforcement of footpath not shown in plan.

MKD	DATE	DESCRIPTION	BY

REVISION

GOVERNMENT OF INDIA
MINISTRY OF SURFACE TRANSPORT
(ROADS WING), NEW DELHI

STANDARD DRAWINGS FOR ROAD BRIDGES

R.C.C. SOLID SLAB
SUPERSTRUCTURE (RIGHT) SPAN 10·0m
(WITH FOOTPATHS)

RECOMMENDED BY		APPROVED BY	1990
(S.S.REDDY) E.E.	(S.K.KAISTHA) S.E.	(P.N.SHIVAPRASAD) C.E.	DRG. NO. SD/122